



Digital Communications

Lecture notes for the whole course: how a waveform becomes a finite alphabet of symbols, how those symbols are carried across a channel that adds noise to everything, how a receiver decides which one was sent, and how often it is wrong.

Covers

Chapters 1 to 6, and appendices A and B

Level

Undergraduate

Assumed background

Fourier analysis, probability, random processes

How to read these notes

Every chapter builds on the one before it. Within a chapter, each idea arrives in the same order: a picture, a definition, an equation, a short derivation, a worked example, and a warning about the mistake that is easiest to make. Worked examples use five headings — Given, Find, Method, Solution, Check. Do the Check step yourself before reading it.

Two conventions apply everywhere and both are worth fixing now, because getting either wrong costs a factor of two that nothing on the page reveals. Noise is white and Gaussian with **two-sided** power spectral density $N_0/2$ watts per hertz. The Gaussian tail is $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2} dt = \frac{1}{2} \operatorname{erfc}(x/\sqrt{2})$. Energy and power are normalised, with $R = 1 \Omega$.

The contents below carries a third column. An entry such as **PS CH7.2.1** points into the course textbook, Proakis and Salehi, *Fundamentals of Communication Systems*, second edition, where the same material is developed at length. The **PS** marker is what tells the two apart, and it is not decorative: these notes reach information theory in chapter 6 and the textbook develops it in chapter 12, while the textbook's own chapter 12 heading is nowhere near what a bare number would suggest.

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|--|--|------------------------------------|
| 1 | The transition from analog to digital | PS CH7.1–7.4 |
| Impulse-train sampling and the replication of the spectrum. The sampling theorem. Reconstruction and sinc interpolation. Uniform quantization, the error it makes, and the signal-to-quantization-noise ratio. Companding. Encoding, line codes and pulse code modulation. | | |
| 2 | Baseband transmission of digital signals | PS CH8.2–8.3, PS CH10.1, PS CH10.3 |
| The matched filter and what it achieves. Correlator and matched-filter demodulators. The decision statistic, the optimal threshold and the bit error probability. Intersymbol interference and the eye pattern. Nyquist's criterion | | |
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and the raised cosine.

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| 3 | Geometric representation of signal waveforms | PS CH8.1 |
| <p>Signals as vectors. Orthonormal bases, coordinates, energy and distance. The constellation diagram. The Gram–Schmidt procedure.</p> | | |
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| 4 | The optimal receiver in additive white Gaussian noise | PS CH8.3–8.4 |
| <p>What the correlator bank keeps. The MAP and ML rules. Minimum-distance detection and the receiver that computes it. Decision regions. The union bound and the nearest-neighbour approximation.</p> | | |
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| 5 | Digital modulation methods | PS CH8.5–8.7, PS CH9.5 |
| <p>The three binary schemes and the three decibels between them. M-ary phase-shift keying and what each extra bit costs. Amplitude-shift keying on a line, quadrature amplitude modulation on a grid, and frequency-shift keying in M dimensions.</p> | | |
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| 6 | An introduction to information theory | PS CH12.1–12.3 |
| <p>Self-information and entropy. Extended sources. Average codeword length and coding efficiency. Uniquely decodable and prefix codes, the Kraft inequality, and how close to the entropy a code can get. Huffman coding, its ties and its variance.</p> | | |
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| A | Summary of formulas | – |
| <p>Everything the course establishes, in the order it establishes it, with no derivations.</p> | | |
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| B | The laboratories | – |
| <p>The four laboratories of the course, each one run rather than described. A uniform quantizer and the ratio it achieves. A matched filter and the bit error rate it makes. Sixteen-point quadrature amplitude modulation against the union bound. Huffman coding and how much it compresses.</p> | | |
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The transition from analog to digital

A continuous waveform becomes a bit stream in three steps: sampling makes it discrete in time, quantization makes it discrete in amplitude, and encoding replaces each amplitude by a word of bits. Only the first of the three can be undone, and this chapter is largely about what that means.

1.1 Impulse-train sampling

Sampling every T_s seconds is multiplication by a train of impulses. Write $f_s = 1/T_s$ for the sampling frequency.

THE IDEAL SAMPLED SIGNAL

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

$$g_\delta(t) = g(t)p(t) = \sum_{n=-\infty}^{\infty} g(nT_s) \delta(t - nT_s)$$

The second line uses the sampling property $g(t)\delta(t - t_0) = g(t_0)\delta(t - t_0)$, which turns the product under the sum into a number.

What makes this worth writing is that g_δ is a continuous-time signal, not a sequence, so it has a Fourier transform. A product in time is a convolution in frequency, and the transform of an impulse train is another impulse train:

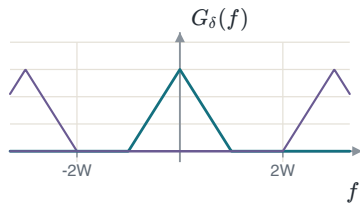
$$P(f) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} \delta(f - nf_s) = f_s \sum_{n=-\infty}^{\infty} \delta(f - nf_s)$$

The coefficient is the Fourier-series coefficient of $p(t)$, and it is the same for every harmonic: $a_k = \frac{1}{T_s} \int_{-T_s/2}^{T_s/2} \delta(t) e^{-j2\pi k f_0 t} dt = \frac{1}{T_s}$, by the sifting property. Convolution with G with that train gives the result the rest of the chapter rests on.

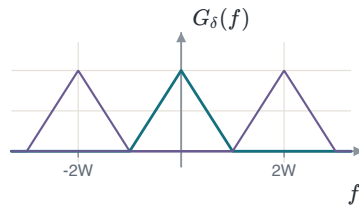
SAMPLING REPLICATES THE SPECTRUM

$$G_\delta(f) = f_s \sum_{n=-\infty}^{\infty} G(f - nf_s)$$

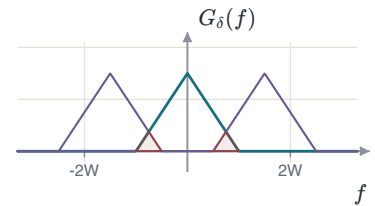
Sampling copies the spectrum to every multiple of f_s and scales it by f_s . Nothing is lost provided the copies do not overlap.



$f_s > 2W$: a gap between the replicas.



$f_s = 2W$: they touch and do not overlap.



$f_s < 2W$: they overlap, and the overlap cannot be undone.

ALIASING

In the third case a high frequency of the message has been added to a low frequency of a replica. No filter can separate a sum into its parts, so the original signal cannot be recovered from its samples — not by a better filter and not by more computation. In practice a real signal is never strictly bandlimited, so a **guard band** f_g is left between the edge of the message and the edge of the first replica: $f_s = 2W + f_g$, with an anti-aliasing filter removing whatever lies above W before the sampler sees it.

1.2 The sampling theorem and reconstruction

SAMPLING THEOREM

If $G(f) = 0$ for $|f| \geq W$ and $f_s \geq 2W$, then $g(t)$ is determined completely by its samples $g(nT_s)$ and can be recovered from them exactly. If $f_s < 2W$, aliasing occurs and the recovery fails. The rate $2W$ is the **Nyquist rate** and $T_s = 1/(2W)$ the **Nyquist interval**.

Recovery means keeping the copy at the origin and rejecting every other one, which is a lowpass filter. At $f_s = 2W$:

THE RECONSTRUCTION FILTER AND ITS IMPULSE RESPONSE

$$H_{\text{LPF}}(f) = \begin{cases} \frac{1}{2W}, & |f| \leq W \\ 0, & \text{otherwise} \end{cases}$$

$$h_{\text{LPF}}(t) = \int_{-W}^W \frac{1}{2W} e^{j2\pi ft} df = \frac{\sin(2\pi Wt)}{2\pi Wt} = \text{sinc}(2Wt)$$

The convention used throughout is $\text{sinc}(x) = \sin(\pi x)/(\pi x)$, so that sinc is one at zero and zero at every other integer.

THE GAIN IS NOT ONE

Sampling multiplied the spectrum by $f_s = 2W$, so the filter has to divide it back. A reconstruction filter of unit gain returns a signal $2W$ times too large, and nothing in a plot of the spectrum *shape* reveals it.

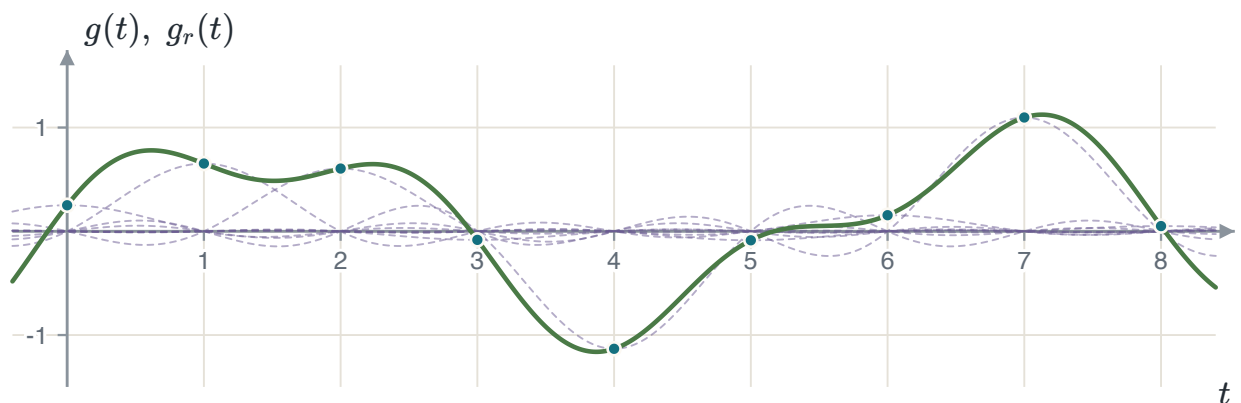
Filtering in frequency is convolution in time. Convolving the impulse train with h_{LPF} and using the sifting property once more gives the interpolation formula.

INTERPOLATION

$$g_r(t) = \sum_{n=-\infty}^{\infty} g(nT_s) \text{sinc}(2W(t - nT_s))$$

$$g_r(t) = \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2W}\right) \text{sinc}(2Wt - n) \quad \text{at } f_s = 2W$$

At $t = kT_s$ every term vanishes except the one with $n = k$, because sinc is zero at every non-zero integer. So $g_r(kT_s) = g(kT_s)$ exactly — and, since the filter passes the whole message and nothing else, the two agree everywhere and not only at the instants.



Each sample contributes one shifted sinc scaled by its own value. Their sum, drawn heavy, passes through every sample because each sinc is zero at all the other sampling instants.

EXAMPLE 1.1 – THREE SAMPLING RATES

GIVEN A signal $x(t)$ bandlimited to $W = 40$ kHz.

FIND (a) the Nyquist rate; (b) the rate with a 10 kHz guard band; (c) the Nyquist rate of $y(t) = x(t) \cos(80000\pi t)$.

METHOD Read (a) and (b) off the geometry of the replicas. For (c) find the bandwidth of y first: the rate follows from that, not from the bandwidth of x .

SOLUTION (a) $f_s = 2W = 80$ kHz. (b) $f_s = 2W + f_g = 90$ kHz. (c) The carrier is at $f_c = 40$ kHz, and multiplication shifts the spectrum both ways and halves it: $Y(f) = \frac{1}{2}X(f - 40\text{k}) + \frac{1}{2}X(f + 40\text{k})$. The result occupies $|f| < 80$ kHz, so $f_s = 160$ kHz.

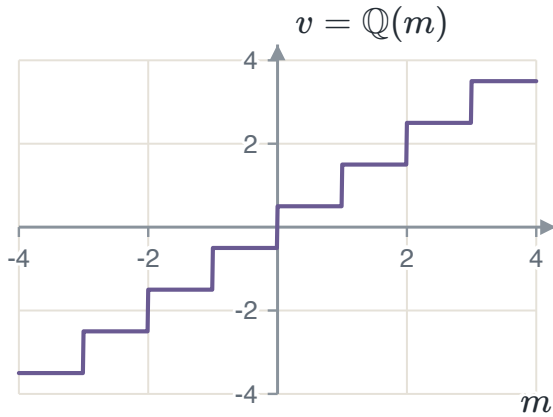
CHECK The highest frequency of y is $f_c + W = 80$ kHz and $2 \times 80 = 160$ kHz. Doubling the carrier would double the answer while leaving the width of Y unchanged: the rate follows the highest frequency present, not the width of the message.

THE TRAP IN PART (C)

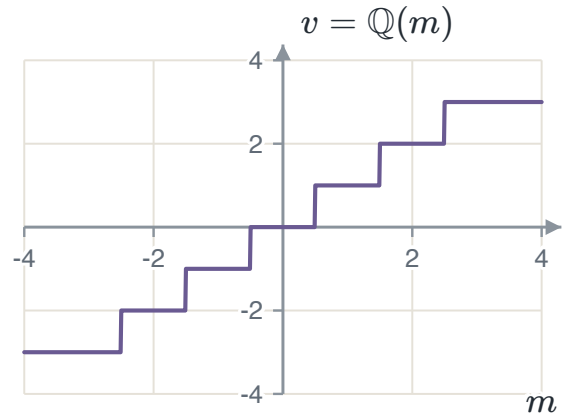
Answering 80 kHz is answering for x , not for y . Modulation moved the message up the frequency axis and the sampler has to keep up with where it went.

1.3 Quantization

Quantization replaces a sample amplitude by the nearest member of a finite set of L **representation levels**. In a **uniform** quantizer the spacing Δ between consecutive levels is the same everywhere. The two families differ in what happens at zero: a **mid-rise** quantizer puts a decision boundary there, so no output level is zero, while a **mid-tread** quantizer puts a level there, so a small input is quantized to exactly zero. Speech coders use mid-tread, because a silent input should be reproduced as silence rather than as a $\pm\Delta/2$ chatter.



Mid-rise, $L = 8$, $\Delta = 1$. A boundary at the origin.



Mid-tread, $L = 8$, $\Delta = 1$. A level at the origin.

A quantizer is a partition of the input range into L regions $\mathcal{J}_k$ with one output value v_k per region. Two conditions make it optimal, and they refer to each other, which is why they are applied by turns rather than solved at once.

1. Each boundary is the **midpoint** of the two levels it separates, $m_k = \frac{1}{2}(v_{k-1} + v_k)$. Given the levels, this minimises the error, because it sends every input to the nearer level.
2. Each level is the **centroid** of its own region, $v_k = E[M \mid M \in \mathcal{J}_k]$. Given the boundaries, this minimises the mean square error inside the region.

A uniform quantizer satisfies the first by construction. It satisfies the second only when the input is uniformly distributed, which is the deeper reason the uniform quantizer is optimal for a uniform source and for no other.

1.4 Quantization noise and the signal-to-noise ratio

The quantization error is $Q = M - \mathbb{Q}(M)$. Because every input is sent to the nearer level, it can never exceed half a step, so $-\Delta/2 \leq q \leq \Delta/2$. If Δ is small enough that the density of M is nearly flat across one region, the error is as likely to fall anywhere in that interval as anywhere else.

THE UNIFORM ERROR MODEL

$$f_Q(q) = \frac{1}{\Delta}, \quad -\frac{\Delta}{2} \leq q \leq \frac{\Delta}{2}$$

$$E[Q^2] = \int_{-\Delta/2}^{\Delta/2} q^2 \frac{1}{\Delta} dq = \frac{\Delta^2}{12}$$

With a zero-mean input the error has zero mean too, so this mean square is also the variance: the power of the quantization noise is its variance.

Substituting $\Delta = 2m_{\max}/L$ and $L = 2^R$ puts the result in the form the rest of the chapter uses, and dividing the signal power by it gives the figure of merit.

SIGNAL-TO-QUANTIZATION-NOISE RATIO

$$E[Q^2] = \frac{1}{12} \left(\frac{2m_{\max}}{L} \right)^2 = \frac{m_{\max}^2}{3 \cdot 2^{2R}}$$

$$\text{SQNR} = \frac{P_M}{E[Q^2]} = \frac{3P_M}{m_{\max}^2} 2^{2R}$$

$$\text{SQNR [dB]} = 10 \log_{10} \underbrace{\frac{3P_M}{m_{\max}^2}}_{\alpha} + 6.02R$$

Every extra bit per sample adds $20 \log_{10} 2 = 6.02$ dB. Doubling the level count halves the step, which quarters the error power, and a factor of four is 6.02 dB.

WHAT α COSTS

The intercept is negative for every signal that does not fill the quantizer range. A full-scale sinusoid has $P_M = m_{\max}^2/2$ and $\alpha = 10 \log_{10} 1.5 \approx 1.76$ dB; a sinusoid reaching a quarter of the range has $\alpha = -10.28$ dB, a loss of 12.04 dB — exactly two bits. That calculation is the argument for companding, and its size is the size of the argument.

EXAMPLE 1.2 — A FULL-SCALE SINUSOID

- GIVEN** $m(t) = 5 \cos t$ through a uniform quantizer spanning its full range.
- FIND** The step size and the SQNR at $R = 3$ and $R = 4$ bits per sample.
- METHOD** Average power from Parseval, step size from $\Delta = 2m_{\max}/L$, then $\alpha + 6.02R$.
- SOLUTION** The coefficients are $a_{\pm 1} = 5/2$, so $P_M = 2(5/2)^2 = 12.5$ and $m_{\max} = 5$. At $R = 3$: $\Delta = 2(5)/8 = 1.25$ V and $\text{SQNR} = 1.76 + 18.06 = 19.82$ dB. At $R = 4$: $\Delta = 0.625$ V and $\text{SQNR} = 1.76 + 24.08 = 25.84$ dB.
- CHECK** The two differ by 6.02 dB, one bit. Independently, $E[Q^2] = \Delta^2/12 = 0.1302$ and $10 \log_{10}(12.5/0.1302) = 19.82$ dB.

WHAT THE FORMULA DOES NOT KNOW

Quantize this waveform and measure the error it actually makes, and the answers are 19.09 dB and 25.31 dB — about 0.7 dB and 0.5 dB below the formula. The uniform model assumes the error is equally likely anywhere in half a step, and a sinusoid spends most of its time near its peaks, where the error is largest. The gap halves with every extra bit: 0.27 dB at $R = 6$ and 0.14 dB at $R = 8$. That is what " Δ small enough" means, and knowing the size of the gap is the difference between using a model and believing it.

EXAMPLE 1.3 — WHEN THE MODEL BREAKS

- GIVEN** A zero-mean stationary Gaussian source with $S_X(f) = 2$ for $|f| < 100$ Hz, sampled at the Nyquist rate, each sample through a five-level quantizer with outputs $-30, -10, 0, 10, 30$ and boundaries at $-40, -20, 20, 40$.
- FIND** The SQNR of the scheme.
- METHOD** The signal power is the area under the spectral density. The noise power must be integrated region by region, because this quantizer is neither uniform nor fine.
- SOLUTION** $P_X = \int_{-100}^{100} 2 df = 400$, and since the mean is zero this is also σ_X^2 . Splitting $P_Q = \int (x - Q(x))^2 f_X(x) dx$ at the four boundaries gives $7.98 + 46.36 + 79.50 + 46.36 + 7.98 = 188.18$, so $\text{SQNR} = 10 \log_{10}(400/188.18) = 3.27$ dB.
- CHECK** The central region alone contributes 79.50, and it holds the 68% of the mass with $|X| < 20$ and an error of up to 20. That is where a five-level quantizer spends its error, and it is why the answer is a few decibels rather than a few tens.

WHY $\Delta^2/12$ WOULD HAVE LIED

The quantizer is coarse and its outer regions are unbounded, so the error there is not bounded by half a step at all: a sample at $x = 120$ is quantized to 30 and the error is 90. Applying the uniform model with $\Delta = 20$ predicts $E[Q^2] = 33.3$ and an SQNR of 10.8 dB, three times too optimistic. The model is a small-step model, and this is what its failure looks like.

1.5 Non-uniform quantization

Speech spends most of its time at small amplitudes and only occasionally reaches its peak. A uniform quantizer gives the same absolute step to a whisper and to a shout, so the whisper is quantized far more coarsely in proportion to itself. Relaxing the requirement that the regions be of equal length allows the distortion to be reduced at the same level count — narrow regions where the density is high and wide ones where it is low, which is what the centroid condition asks for and what a uniform quantizer cannot give.

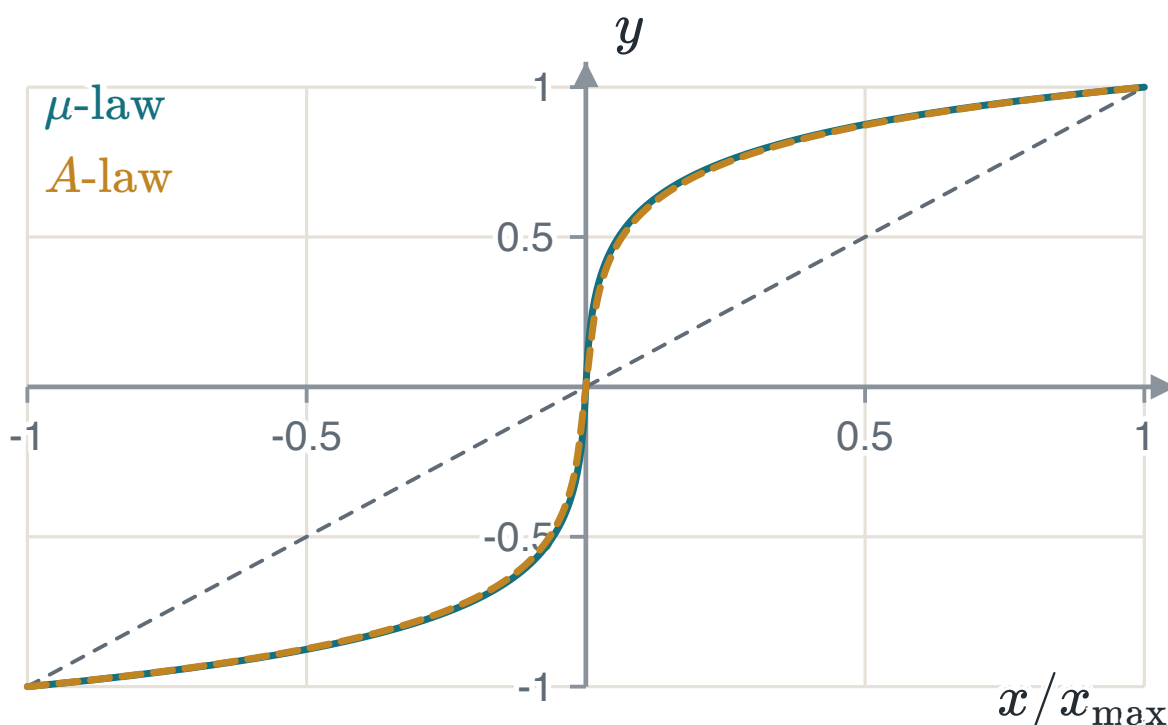
Rather than build a non-uniform quantizer, the signal is compressed by a memoryless non-linearity, quantized **uniformly**, and expanded at the receiver by the inverse. Compressing plus expanding is **companding**, and it is how the non-uniform quantizer is built out of a uniform one.

THE TWO STANDARD COMPRESSORS

$$y = \frac{\ln(1 + \mu|x|)}{\ln(1 + \mu)} \operatorname{sgn}(x), \quad |x| \leq 1$$

$$y = \begin{cases} \frac{A|x|}{1 + \ln A} \operatorname{sgn}(x), & 0 \leq |x| \leq 1/A \\ \frac{1 + \ln(A|x|)}{1 + \ln A} \operatorname{sgn}(x), & 1/A < |x| \leq 1 \end{cases}$$

Both are normalised so that $x = \pm 1$ maps to $y = \pm 1$. The standard parameters are $\mu = 255$ and $A = 87.6$; on the same speech through the same eight-bit uniform quantizer the two give signal-to-noise ratios within a hundredth of a decibel of one another. The difference between them is regional rather than technical.

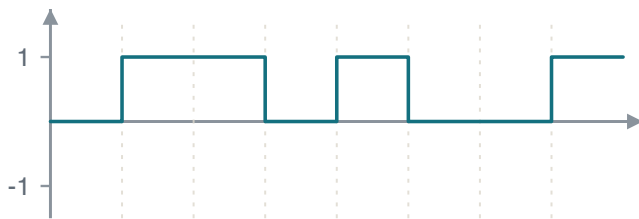


The two compressor characteristics against the identity, which is what no companding would give. Both spend most of their output range on the smallest tenth of the input.

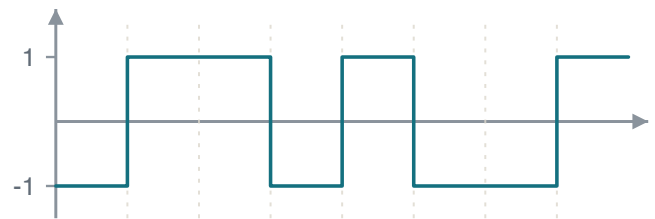
1.6 Encoding, line codes and pulse code modulation

With $L = 2^R$ levels each sample needs R bits, and at f_s samples per second the stream leaves the encoder at $R_b = Rf_s$ bits per second. **Natural binary coding** assigns 0 to $L - 1$ to the levels in increasing order; **Gray coding** assigns the words so that adjacent levels differ in exactly one bit. The second is worth the trouble because a channel error almost always moves the decision to a neighbouring level, and under natural binary that can flip several bits at once — level 7 is 0111 and level 8 is 1000, four bits apart, where the Gray words 0100 and 1100 are one.

A **line code** is the rule that turns those bits into a waveform. Unipolar NRZ is on-off signalling: simplest, and it carries a DC level that makes the waveform droop through any AC-coupled stage. Polar NRZ is antipodal signalling and has no DC component for balanced data, which is why it is the shape Chapter 2 analyses. Both go quiet during a long run of identical bits, and a receiver recovering its clock from the waveform then has nothing to lock to; Manchester forces a transition in the middle of every bit, so the clock is always recoverable and there is never a DC component, paid for with twice the bandwidth.



Unipolar NRZ.



Polar NRZ, carrying the same eight bits 01101001.

EXAMPLE 1.4 – A COMPLETE PCM STREAM

- GIVEN** $m(t) = 8|\text{sinc}(t - 2)|$, so $0 \leq m(t) \leq 8$, sampled every $T_s = 0.6$ s and quantized by an eight-level uniform quantizer covering $[0, 8]$.
- FIND** The step size, the levels, the code words for $t = 0, 0.6, \dots, 3.6$, and the bit rate.
- METHOD** Δ from the range and the level count; each sample by direct evaluation; each word from the tread the sample falls in.
- SOLUTION** $\Delta = 8/8 = 1$ V and the levels sit at 0.5, 1.5, $\dots$, 7.5. The samples are 0, 1.73, 1.87, 7.48, 6.05, 0, 1.51, giving the words 000 001 001 111 110 000 001. With $R = 3$ and $f_s = 1/0.6 = 1.667$, $R_b = 5$ bit/s and $T_b = T_s/3 = 0.2$ s.
- CHECK** The sample at $t = 3$ is $8|\text{sinc}(1)| = 0$ exactly, because sinc vanishes at every non-zero integer — the same fact the interpolation formula rests on, met again at the other end of the chapter.
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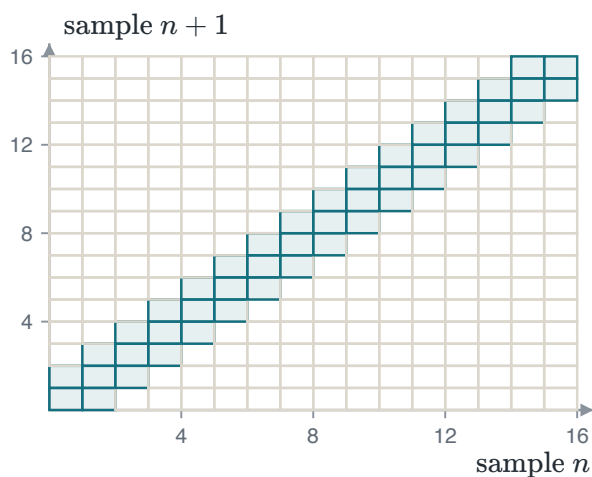
1.7 Vector quantization

Every quantizer so far has rounded one sample on its own. That is **scalar** quantization, and it throws something away before it starts: real signals do not jump about from one sample to the next, and a quantizer looking at one sample at a time cannot know that. **Vector quantization** takes n samples at a time, treats them as one point in n dimensions, and rounds the whole block to the nearest entry of a

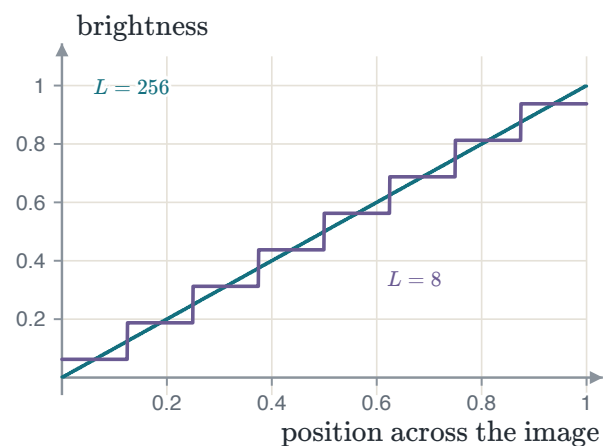
codebook. With $n = 1$ it is the quantizer of section 1.3; with $n = 2$ the codebook entries are places in a plane, and they can be put wherever the signal actually goes.

EXAMPLE 1.5 – PAIRS OF NEIGHBOURING SAMPLES

- GIVEN** $L = 16$ levels, and a signal smooth enough that a sample never moves more than one step from the one before it.
- FIND** What a pair costs, quantized separately and quantized together.
- METHOD** Count the pairs each scheme has to be able to name, and take the base-two logarithm.
- SOLUTION** Separately, every combination is allowed: $L^2 = 256$ pairs, $\log_2 256 = 8$ bits a pair, or 4 bits a sample. Together, only the pairs with $|i - j| \leq 1$ can occur, and there are $3L - 2 = 46$ of them: $\lceil \log_2 46 \rceil = 6$ bits a pair, or 3 bits a sample.
- CHECK** The cells did not change size, so the rounding error did not change either — the saving is a quarter of the rate at the same distortion. What was dropped is the cost of naming the 210 combinations the signal never produces. That is the whole idea: **spend the codebook where the signal goes.**



Every pair a 16-level quantizer can produce. A scalar quantizer pays for all 256 squares; if a sample never moves by more than one step, only the 46 shaded ones ever occur.



One row of a smooth gradient at 256 levels and at 8. The coarse staircase is banding: each step edge is a boundary the eye reads as a line that was not in the scene.

THE GAIN IS IN THE DEPENDENCE, MOSTLY

Almost all of the saving comes from neighbouring samples being alike. A small gain survives even for independent samples, because better-shaped cells pack the space more efficiently than squares, but it is under a quarter of a bit a sample and this course does not pursue it. For a source that really is independent and evenly spread, scalar quantization is close to the best there is.

EXAMPLE 1.6 – QUANTIZING AN IMAGE

GIVEN	A 512×512 greyscale image at 8 bits a pixel, so $L = 256$ levels.
FIND	The size of the file, and the size and the cost at $L = 32$.
SOLUTION	$512^2(8) = 2\,097\,152$ bits, which is 256 KiB. At $L = 32$ the rate is $R = \log_2 32 = 5$ bits a pixel, so 1 310 720 bits or 160 KiB — 37.5% smaller.
WHAT IT COSTS	$\Delta\text{SQNR} = 6.02(8 - 5) = 18.06$ dB, and it is visible as well as measurable: a smooth gradient becomes a staircase, and the picture shows flat bands with hard edges where the scene had none.
CHECK	Nothing here is recoverable. The bits dropped described where inside its level each pixel really sat, and no decoder invents them back. That is what lossy names, and it is the trade every image format makes — JPEG among them, which quantizes a transform of each block rather than the pixels themselves, but quantizes all the same.

1.8 Summary

RESULT	STATEMENT	ANCHOR
Replication	$G_\delta(f) = f_s \sum_n G(f - nf_s)$	PS CH7.1.1
Sampling theorem	$f_s \geq 2W$ for a message bandlimited to W	PS CH7.1.1
Reconstruction	$g_r(t) = \sum_n g(nT_s) \text{sinc}(2Wt - n)$, filter gain $1/(2W)$	PS CH7.1.1
Step size	$\Delta = 2m_{\max}/L$, $L = 2^R$	PS CH7.2.1
Error power	$E[Q^2] = \Delta^2/12$, when the step is small	PS CH7.2.1
Signal-to-noise	$\text{SQNR} [\text{dB}] = \alpha + 6.02R$	PS CH7.2.1
Bit rate	$R_b = Rf_s$	PS CH7.3, 7.4.1
Vector quantization	round n samples together; spend the codebook where the signal goes	PS CH7.2.2

Sampling is reversible and quantization is not. Everything after this chapter takes the bit stream as given and asks what the channel does to it.

CHAPTER 2

Baseband transmission of digital signals

Chapter 1 produced a stream of bits. Now each bit is sent as a waveform, a channel adds noise to it, and a receiver has to say which waveform was sent. This chapter answers three questions in order: what filter the receiver should use, where the decision threshold should go, and what happens when the channel is too narrow and the pulses start to overlap.

2.1 The matched filter

Over one bit interval the receiver sees $x(t) = g(t) + w(t)$, the waveform plus white Gaussian noise of two-sided density $N_0/2$. It passes this through a filter and takes one sample at the end of the interval.

Because the filter is linear, its output splits the same way, into a signal part $g_0(t) = g * h$ and a noise part $n(t) = w * h$.

The quantity worth maximising is the signal at the sampling instant against the noise power that accompanies it.

THE PEAK PULSE SIGNAL-TO-NOISE RATIO

$$(\text{SNR})_o = \frac{|g_0(T)|^2}{E[n^2(t)]} = \frac{|\int_{-\infty}^{\infty} G(f)H(f)e^{j2\pi fT}df|^2}{\frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 df}$$

Scaling H up multiplies the top and the bottom by the same factor, so the answer cannot be "make H large". What is being chosen is the shape of H .

Schwarz's inequality says that $|\int \phi_1 \phi_2|^2 \leq \int |\phi_1|^2 \int |\phi_2|^2$, with equality only when $\phi_1 = k\phi_2^*$. Taking $\phi_1 = H(f)$ and $\phi_2 = G(f)e^{j2\pi fT}$, the factor $\int |H|^2$ cancels between the top and the bottom and a bound appears that contains no H at all.

THE BOUND, AND THE FILTER THAT REACHES IT

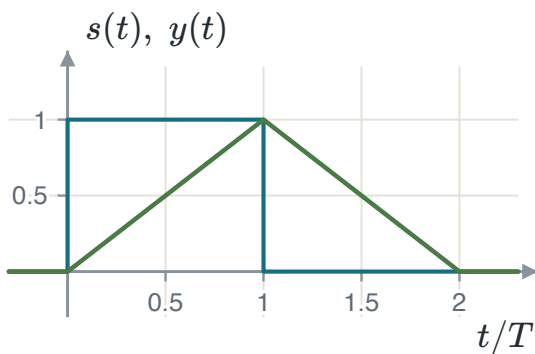
$$(\text{SNR})_o \leq \frac{2}{N_0} \int |G(f)|^2 df = \frac{2E}{N_0}$$

$$H(f) = k G^*(f) e^{-j2\pi fT} \iff h_{\text{opt}}(t) = k g(T - t)$$

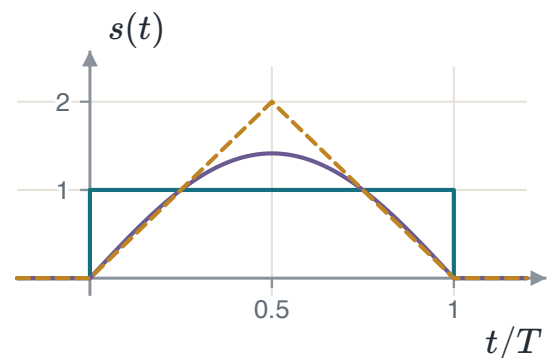
The optimum filter is the transmitted waveform reversed in time and shifted into the interval. It is called the **matched filter**.

TWO THINGS THIS SAYS

The best achievable ratio is $2E/N_0$: it depends on the **energy** of the pulse and on the noise density, and on nothing else about the pulse. Two completely different waveforms of the same energy perform identically. And the constant k cancels, so the filter is fixed only up to a gain.



A rectangular pulse and the output of the filter matched to it. The output peaks at exactly $t = T$, and its peak is the energy of the pulse.



Three pulses of the same energy. Against white Gaussian noise a matched-filter receiver performs identically on all three.

2.2 From waveform to number

Both waveforms of a binary baseband system are multiples of one shape, so a single unit-energy function carries both. With $\psi(t) = 1/\sqrt{T_b}$ on the bit interval, polar NRZ is $s_m(t) = s_m\psi(t)$ with $s_0 = -A\sqrt{T_b}$ and $s_1 = +A\sqrt{T_b}$. Both carry energy $E_b = A^2T_b$, so $A\sqrt{T_b} = \sqrt{E_b}$ and the two waveforms have become two *numbers*, $\pm\sqrt{E_b}$, on one axis.

The demodulator can be built two ways. The **matched filter** convolves with $\psi(T_b - t)$ and samples at T_b . The **correlator** multiplies by $\psi(t)$ and integrates over the interval. At the sampling instant the two produce the same number, because convolving with a reversed function and evaluating at the end of the interval is correlating over it.

THE DECISION STATISTIC

$$y = \int_0^{T_b} x(\tau)\psi(\tau) d\tau = s_m \underbrace{\int_0^{T_b} \psi^2}_{=1} + \underbrace{\int_0^{T_b} w\psi}_{n} = s_m + n$$

They agree at $t = T_b$ and nowhere else, which is one more reason the sampling instant matters.

2.3 The decision and its error probability

The noise term is a projection of a Gaussian process onto a fixed function, so it is a Gaussian random variable. Its variance follows from $E[w(\tau)w(u)] = \frac{N_0}{2}\delta(\tau - u)$ and the sifting property, and the unit energy of ψ makes the answer independent of the interval length.

WHAT THE DETECTOR IS GIVEN

$$\sigma_n^2 = \frac{N_0}{2} \int_0^{T_b} \psi^2(\tau) d\tau = \frac{N_0}{2}$$

$$y = s_m + n \sim \mathcal{N}\left(s_m, \frac{N_0}{2}\right)$$

The two-sided convention arrives here unchanged. Reading the density as one-sided makes every error probability in the course 3 dB too optimistic, and nothing in the algebra shows it.

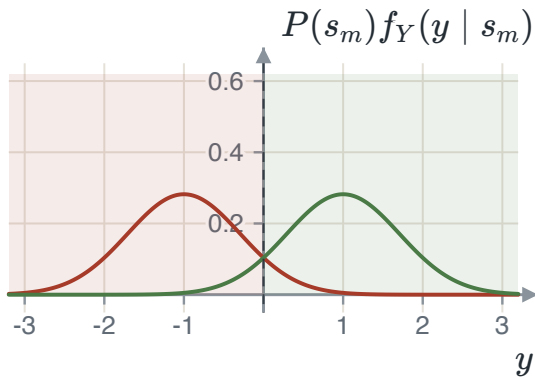
The detector decides "1" when $y > \lambda$. Writing the average error as a sum of two integrals, differentiating with respect to λ by the Leibniz rule and setting the derivative to zero gives a condition that is easy to read: the threshold sits where the two densities, each weighted by its prior, cross.

THE OPTIMAL THRESHOLD

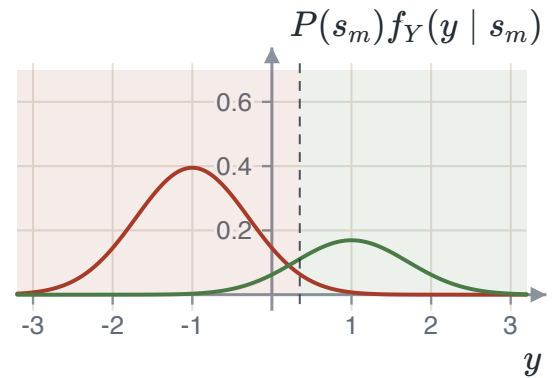
$$P(s_1)f_Y(\lambda | s_1) = P(s_0)f_Y(\lambda | s_0)$$

$$\lambda_{\text{opt}} = \frac{N_0}{4\sqrt{E_b}} \ln \frac{P(s_0)}{P(s_1)}$$

Equal priors put it midway. If s_0 is the more likely symbol the threshold moves in the positive direction, enlarging the region that decides s_0 — which is the right way round.



Equal priors: the threshold sits midway and the shaded decision regions are symmetric.



$P(s_0) = 0.7$: the more likely symbol's curve is taller and the crossing has moved towards the less likely one.

With equal priors the threshold is at the origin and both conditional errors are the same Gaussian tail, using $P(Y > y) = Q\left(\frac{y-\mu}{\sigma}\right)$ and $Q(-x) = 1 - Q(x)$.

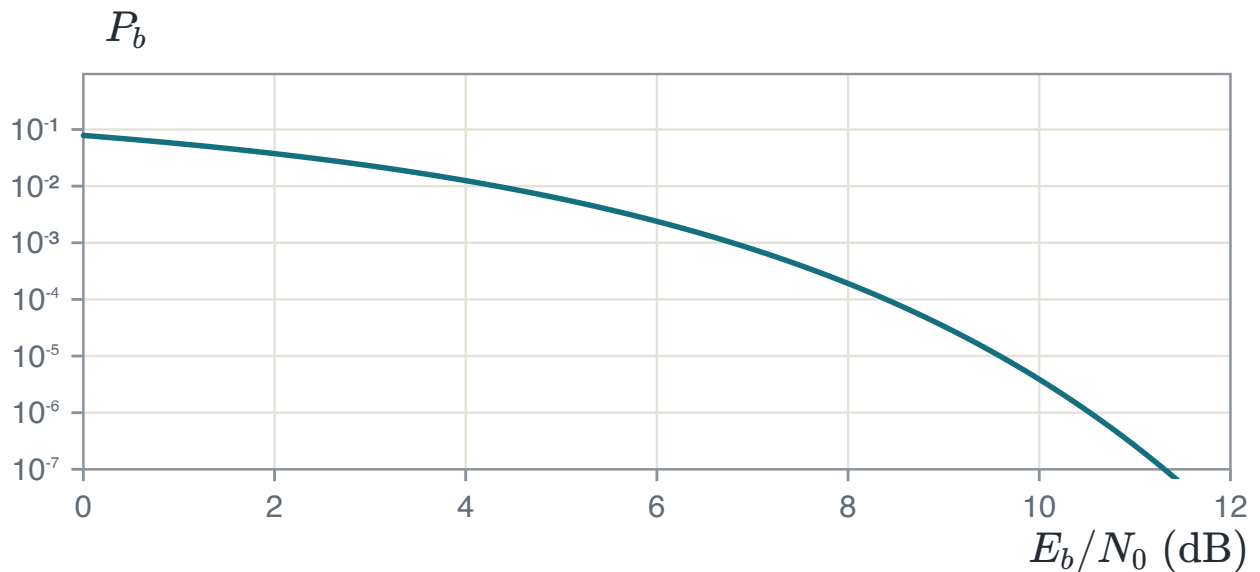
THE RESULT OF THE CHAPTER

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

It depends on E_b/N_0 and on nothing else — not on the amplitude, not on the bit duration, not on the pulse shape. Doubling the amplitude and quartering the duration change neither the energy per bit nor the answer.

THE FACTOR OF TWO THAT LIVES HERE

$Q\left(\sqrt{E_b/N_0}\right)$, the same expression without the two, is the error probability of on-off or orthogonal signalling, and it is 3 dB worse. Which one applies depends on whether the two waveforms are opposites or merely different. For on-off signalling half the bits carry no energy at all, so the average energy per bit is half the peak — and forgetting that is how on-off comes out looking as good as antipodal.



Bit error probability against E_b/N_0 . Past about 8 dB every extra decibel is worth roughly an order of magnitude in error rate.

EXAMPLE 2.1 – UNEQUAL PRIORS

- GIVEN** A binary PAM system with correlator output $y = \pm\sqrt{E_b} + n$, where $P(s_1) = 0.3$, $E_b = 1$ and $N_0 = 0.1$.
- FIND** The optimal threshold and the average error probability there.
- METHOD** Threshold from the log-ratio of the priors; each conditional error from a Gaussian tail; average by weighting.
- SOLUTION** $\lambda = \frac{0.1}{4} \ln \frac{0.7}{0.3} = 0.0212$ and $\sigma = \sqrt{0.05} = 0.2236$. Then $P(\text{err} | s_0) = Q(4.567) = 2.475 \times 10^{-6}$ and $P(\text{err} | s_1) = Q(4.377) = 6.005 \times 10^{-6}$, so $P_e = 0.7(2.475) + 0.3(6.005)$ in units of 10^{-6} , giving 3.534×10^{-6} .
- CHECK** Leaving the threshold at zero would give $Q(\sqrt{20}) = 3.872 \times 10^{-6}$. Moving it improves the answer by about nine per cent — a real gain and a small one, which is what a shift of a tenth of a standard deviation buys.
-

2.4 Intersymbol interference

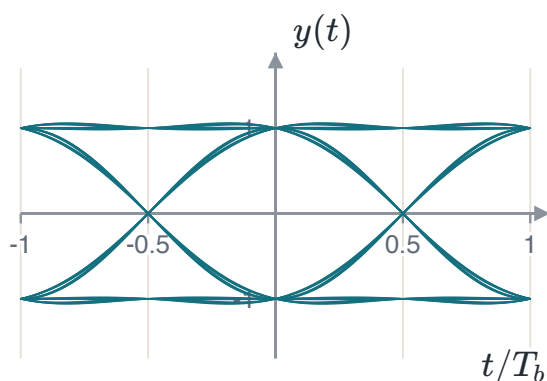
Everything above assumed an ideal channel. A real channel is bandlimited, and a bandlimited channel spreads each pulse in time, so a pulse meant for one interval leaks into the next. Writing the received signal as a train of the overall pulse p and sampling at $t_i = iT_b$ separates the term that was meant to be there from the ones that were not.

WHERE THE INTERFERENCE COMES FROM

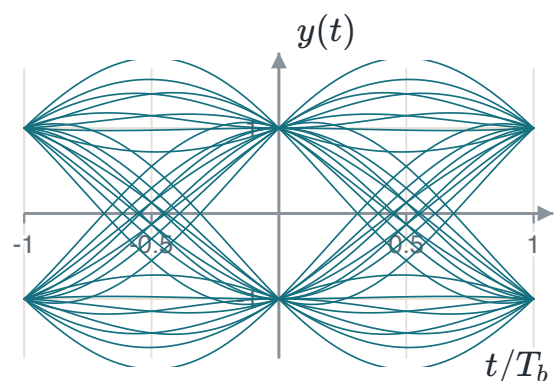
$$y(t_i) = \underbrace{\mu a_i}_{\text{wanted}} + \underbrace{\mu \sum_{k \neq i} a_k p((i-k)T_b)}_{\text{intersymbol interference}} + n(t_i)$$

The middle term is not noise. It is caused by the data itself, and raising the transmit power raises it by the same factor, so more power does not help. At high signal-to-noise ratio it is the only thing limiting the system.

The **eye pattern** is how this is seen rather than calculated. Cut the received waveform into bit-length pieces, draw them all on top of each other synchronised to the clock, and four measurements are readable at once: the height of the opening is the margin over noise, its width is how far the sampling instant may drift, the slope at the crossings is the sensitivity to a timing error, and the spread of the crossings is the timing jitter. When the opening closes there is no sampling instant at which every bit pattern decides correctly, and no threshold placement helps.



A wide-open eye: a pulse that decays quickly. Every trace passes close to ± 1 at the centre.



A slowly decaying pulse. The opening has narrowed and the crossings have scattered, with the noise unchanged.

2.5 Nyquist's criterion and the raised cosine

The interference vanishes exactly when the overall pulse is zero at every sampling instant but its own. Sampling p at those instants and transforming turns that requirement into a statement about the spectrum.

NYQUIST'S CRITERION FOR DISTORTIONLESS TRANSMISSION

$$p((i - k)T_b) = \begin{cases} 1, & i = k \\ 0, & i \neq k \end{cases}$$

$$R_b \sum_n P(f - nR_b) = 1, \quad R_b = \frac{1}{T_b}$$

In words: the replicas of the pulse spectrum, spaced by the symbol rate, must add to a constant. The simplest spectrum that does it is a rectangle of width $2W$ with $W = R_b/2$, the **Nyquist bandwidth**, whose pulse is $\text{sinc}(2Wt)$.

THE RATE A BANDWIDTH SUPPORTS

A channel of bandwidth W carries at most $2W$ symbols per second with no interference. This is the counterpart of the sampling theorem, reached from the other end of the same argument.

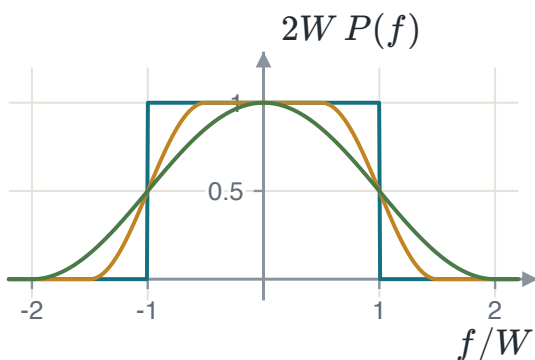
The ideal Nyquist channel cannot be built: its spectrum has vertical edges, and its pulse decays only as $1/t$, so a small timing error lets a long tail of neighbours contribute at once. The fix is to widen the spectrum and taper its edges while keeping the tiling property.

THE RAISED COSINE

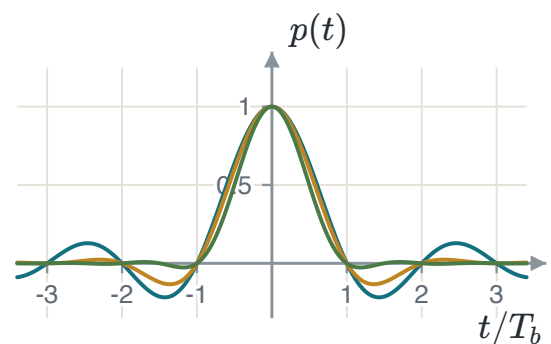
$$P(f) = \begin{cases} \frac{1}{2W}, & 0 \leq |f| \leq f_1 \\ \frac{1}{4W} \left[1 - \sin \frac{\pi(|f| - W)}{2W - 2f_1} \right], & f_1 \leq |f| < 2W - f_1 \\ 0, & \text{otherwise} \end{cases}$$

$$p(t) = \text{sinc}(2Wt) \frac{\cos(2\pi\alpha Wt)}{1 - 16\alpha^2 W^2 t^2}, \quad \alpha = 1 - \frac{f_1}{W}$$

The first factor keeps the zero crossings at $t = iT_b$, so the criterion still holds; the second decays as $1/t^2$. The price is bandwidth: $B_T = (1 + \alpha)W$, an excess of αW over Nyquist.



The spectrum at $\alpha = 0, 0.5$ and 1 . All three tile the axis at spacing $2W$, so all three give zero interference.



Their pulses. All three vanish at every non-zero multiple of T_b ; the tails differ by orders of magnitude.

EXAMPLE 2.2 – FITTING A RATE INTO A CHANNEL

GIVEN	A channel of bandwidth 48 kHz is to carry 64 kbit/s with raised-cosine shaping.
FIND	The Nyquist bandwidth and the largest roll-off that fits.
METHOD	$W = R_b/2$, then $(1 + \alpha)W \leq B$.
SOLUTION	$W = 32$ kHz, and $(1 + \alpha)(32) \leq 48$ gives $\alpha \leq 0.5$. At $\alpha = 0.5$ the signal uses the whole 48 kHz, with an excess of 16 kHz over the Nyquist bandwidth.
CHECK	At $\alpha = 0$ the signal would need only 32 kHz and would also fit — and would be a bad choice, because its pulse decays as $1/t$ and its filter cannot be built. Using the whole channel buys a pulse decaying as $1/t^3$ and a filter that exists. The spectral efficiency is $64/48 = 1.33$ bit/s/Hz against the theoretical 2.

2.6 Summary

RESULT	STATEMENT	ANCHOR
Matched filter	$h_{\text{opt}}(t) = g(T - t)$	PS CH8.3.2
What it achieves	$(\text{SNR})_o = 2E/N_0$, independent of the pulse shape	PS CH8.3.2
Decision statistic	$y = s_m + n$, $n \sim \mathcal{N}(0, N_0/2)$	PS CH8.3.1
Optimal threshold	$\lambda = \frac{N_0}{4\sqrt{E_b}} \ln \frac{P(s_0)}{P(s_1)}$	PS CH8.3.3
Antipodal error	$P_b = Q\left(\sqrt{2E_b/N_0}\right)$	PS CH8.3.3
On-off error	$P_b = Q\left(\sqrt{E_b/N_0}\right)$, three decibels worse	PS CH8.3.3
Nyquist criterion	$R_b \sum_n P(f - nR_b) = 1$	PS CH10.3.1
Raised cosine	$B_T = (1 + \alpha)W$	PS CH10.3.1

Two waveforms have become two points on a line, and the error probability has turned out to depend on the distance between them. Chapter 3 asks what happens when there are more than two waveforms, and finds that the same picture works with more axes.

CHAPTER 3

Geometric representation of signal waveforms

This chapter changes the language, not the subject. A set of waveforms is written as a set of points. Once that is done, the receiver becomes a question about geometry, and the answers are arithmetic instead of integrals.

3.1 Why one axis is not enough

In Chapter 2 the two waveforms were opposites. One shape carried both, the receiver produced one number, and everything followed from that. Now suppose the two waveforms are simply different: neither

is a multiple of the other. No single function can write both as a multiple of itself, so the receiver of Chapter 2 does not apply.

The fix is to use two matched filters, one for each waveform, and let the receiver work with the two numbers they produce. A pair of numbers is a point in a plane. That plane is the subject of this chapter.

3.2 Signals as vectors

A vector is a list of numbers because it has been written against a set of axes. A signal can be a list of numbers for the same reason. The only thing needed is an inner product, and for signals it is the integral of the product.

INNER PRODUCT, ENERGY AND ORTHOGONALITY

$$\langle x, y \rangle = \int_{-\infty}^{\infty} x(t) y(t) dt$$

$$\int_{-\infty}^{\infty} \psi_j(t) \psi_k(t) dt = \begin{cases} 1, & j = k \\ 0, & j \neq k \end{cases}$$

The second line defines an **orthonormal** set. The first case says each function has unit energy; the second says any two of them are orthogonal.

WHAT ORTHOGONAL MEANS HERE

The same thing it means for arrows: the two signals have nothing of each other in them. Multiply them together, add up the result over all time, and you get zero. Two pulses that never overlap are the clearest case — where one is non-zero the other is zero, so the product is zero everywhere and no integration is needed.

With the axes fixed, a signal is written against them the same way a vector is. One integral per axis takes the waveform apart; adding the pieces back builds it again.

COORDINATES

$$s_{ij} = \int_0^T s_i(t) \psi_j(t) dt$$

$$s_i(t) = \sum_{j=1}^N s_{ij} \psi_j(t)$$

The list $\mathbf{s}_i = (s_{i1}, \dots, s_{iN})$ is the **signal vector**.

The reason this is worth doing is that inner products survive the translation. Expand both signals in the basis, exchange the integral with the sums, and use orthonormality: every cross term vanishes and every matching term contributes one. An integral over waveforms becomes a sum over N numbers.

$$\int_{-\infty}^{\infty} x(t)y(t) dt = \sum_{k=1}^N x_k y_k$$

$$E_{s_i} = \|\mathbf{s}_i\|^2 = \sum_{j=1}^N s_{ij}^2, \quad \|\mathbf{s}_i - \mathbf{s}_k\|^2 = \int_0^T (s_i - s_k)^2 dt$$

Energy is how far a point is from the origin. That is what the transmitter pays for. **Distance is how far two points are from each other.** That is what decides how often the receiver confuses them.

3.3 The constellation diagram

The constellation is the picture of the signal vectors in the space their basis spans: one point per waveform, one axis per basis function. Three things are read straight off it. The distance from the origin to a point is the square root of that signal's energy. The distance between two points is the square root of the energy of their difference. And the number of axes is the number of matched filters the receiver needs.

TWO STATEMENTS, AND WHAT THEY MEAN TOGETHER

Two entirely different signal sets can have the same geometric representation. Different shapes, different durations, the same points. And **the geometry determines the performance and the receiver structure.** Put together: if two signal sets give the same constellation, they have the same error probability and the same receiver, and nothing about the waveforms themselves survives into the answer. What the picture does *not* fix is bandwidth, and that is decided by the criteria of Chapter 2.

3.4 The Gram–Schmidt procedure

The axes have been assumed so far. Gram–Schmidt produces them from any set of waveforms. It is one step repeated: take the next signal, remove the part of it that the axes already found can account for, and make an axis out of what is left.

THE PROCEDURE

$$\psi_1(t) = \frac{s_1(t)}{\sqrt{E_1}}, \quad s_{11} = \sqrt{E_1}$$

$$g_k(t) = s_k(t) - \sum_{i=1}^{k-1} s_{ki} \psi_i(t), \quad \psi_k(t) = \frac{g_k(t)}{\sqrt{E_{g_k}}}$$

If $g_k(t) = 0$ the signal was already a combination of the axes found so far and **no new axis is added**. The number of axes N is therefore at most the number of signals M , and is often fewer.

THE ORDER CHANGES THE AXES, NOT THE ANSWER

Starting from a different signal gives a different basis. It gives the same number of axes, the same energies and the same distances, so the constellation is the same picture seen from a different angle. Every result that depends only on distances is unchanged — and in Chapter 4 every result depends only on distances.

EXAMPLE 3.1 – THREE PULSES, TWO AXES

GIVEN $s_1(t) = 1$ on $[0, 2]$; $s_2(t) = 1$ on $[2, 3]$; $s_3(t) = 1$ on $[0, 3]$; each zero elsewhere.

FIND An orthonormal basis, the three signal vectors, and the constellation.

METHOD Normalise the first signal. Then, for each later signal, subtract the part that lies along the axes already found and normalise the remainder.

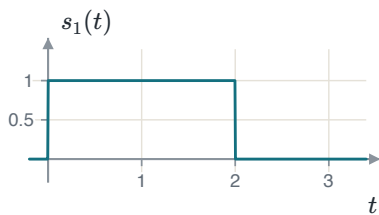
STEP 1 $E_1 = \int_0^2 1 dt = 2$, so $\psi_1(t) = 1/\sqrt{2}$ on $[0, 2]$ and $s_{11} = \sqrt{2}$.

STEP 2 $s_{21} = \int s_2 \psi_1 dt = 0$, because the two never overlap. So $g_2 = s_2$, its energy is 1, and $\psi_2(t) = s_2(t)$.

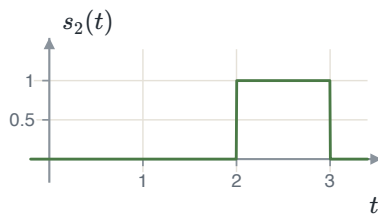
STEP 3 $s_{31} = \sqrt{2}$ and $s_{32} = 1$, and then $g_3 = s_3 - \sqrt{2}\psi_1 - \psi_2 = 0$. No third axis is added.

VECTORS $\mathbf{s}_1 = (\sqrt{2}, 0)$, $\mathbf{s}_2 = (0, 1)$, $\mathbf{s}_3 = (\sqrt{2}, 1)$.

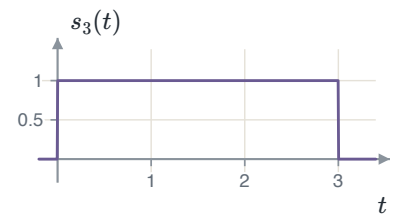
CHECK Energies from the vectors are 2, 1 and 3. From the waveforms: height one for two seconds, one second and three seconds, giving 2, 1 and 3. They agree, and that agreement is the whole content of "energy is squared length". Three waveforms needed only two axes, because $s_3 = s_1 + s_2$ — visible in the pictures before any integral is computed.



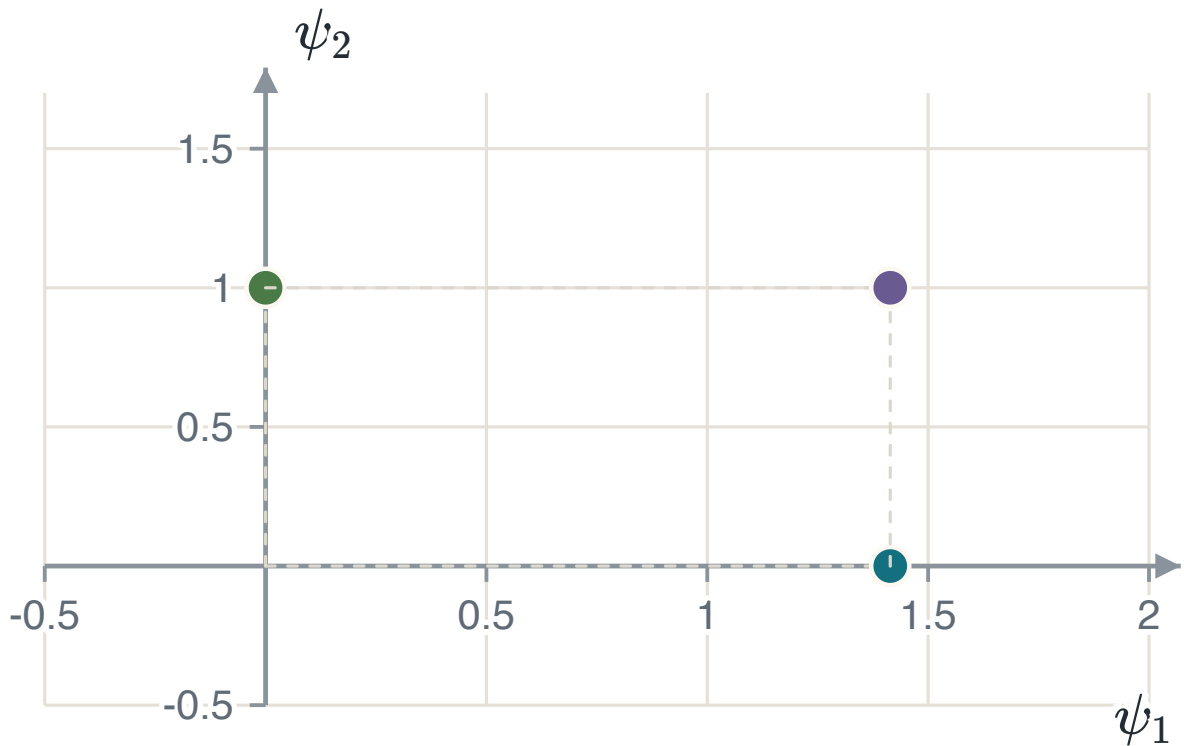
s_1



s_2



$s_3 = s_1 + s_2$



The constellation. Two axes carry three signals. The point furthest from the origin is s_3 , whose energy is 3; the two nearest points are s_1 and s_3 , one unit apart.

3.5 Summary

RESULT	STATEMENT	ANCHOR
Orthonormal set	$\int \psi_j \psi_k dt = 1 \text{ if } j = k, \text{ else } 0$	PS CH8.1
Coordinates	$s_{ij} = \int s_i \psi_j dt$	PS CH8.1
Inner product survives	$\int xy dt = \sum_k x_k y_k$	PS CH8.1
Energy	$E_i = \ s_i\ ^2$	PS CH8.1
Distance	$\ s_i - s_k\ ^2 = \text{energy of the difference}$	PS CH8.1
Gram–Schmidt	$g_k = s_k - \sum_{i < k} s_{ki} \psi_i$, and $\psi_k = g_k / \sqrt{E_{g_k}}$	PS CH8.1

Chapter 4 builds the receiver for this picture. It computes the coordinates of whatever arrives and picks the nearest point, and the error probability then depends on the distances between the points and on nothing else.

CHAPTER 4

The optimal receiver in additive white Gaussian noise

A transmitter picks one of M signals and sends it; the channel adds noise; the receiver has to name the signal. Chapter 3 turned the signals into points. This chapter turns the receiver into a rule about those points, and the rule is simple: choose the point nearest to what arrived.

4.1 What the receiver keeps

The received signal is $r(t) = s_i(t) + n(t)$ over one symbol interval. The receiver has N correlators, one for each basis function, and each returns one number.

THE OBSERVATION VECTOR

$$r_k = \int_0^T r(t) \psi_k(t) dt = s_{ik} + n_k$$
$$\mathbf{r} = \mathbf{s}_i + \mathbf{n} = (r_1, \dots, r_N)$$

The signal was built from the N basis functions, so the correlators capture all of it. The noise was not: the part of it outside the signal space, written $n_0(t)$, is thrown away.

WHY THROWING $n_0(t)$ AWAY COSTS NOTHING

It contains no signal — every signal is a combination of the N basis functions and has no component outside that space. And it is independent of the N numbers the receiver kept. Something that carries no information about the answer and is unrelated to what was kept cannot help. Note what this does *not* say: the noise inside the signal space is kept in full, and it is all the trouble there is.

Each n_k is a projection of a Gaussian process onto a fixed function, so it is Gaussian with zero mean. The useful part is what happens between two of them: using $E[n(\tau)n(u)] = \frac{N_0}{2} \delta(\tau - u)$ and orthonormality, $E[n_j n_k]$ is $N_0/2$ when $j = k$ and zero otherwise. Uncorrelated jointly Gaussian variables are independent, so the noise components are independent, each $\mathcal{N}(0, N_0/2)$.

WHAT THAT MEANS FOR THE PICTURE

Every axis carries the same variance and the axes are independent, so the noise cloud is a **circle** — a sphere in N dimensions — with no preferred direction. That symmetry is why the receiver may simply measure distance. If the variances differed, the nearest point in the ordinary sense would not be the best answer.

4.2 The decision rule

The receiver should choose the signal that is most likely given what it saw. By Bayes' rule that means maximising $P(\mathbf{s}_i) f_{\mathbf{r}}(\mathbf{r} | \mathbf{s}_i)$ — the prior times the likelihood. This is the **MAP** rule, and no rule has a smaller error probability. If all M signals are equally likely the priors cannot change the answer, and what is left is the **ML** rule: maximise the likelihood alone.

Putting the noise density into the likelihood and taking a logarithm — which is increasing, so it changes no winner — turns the product into a sum and leaves one term that depends on i .

MINIMUM-DISTANCE DETECTION

$$\ln f_{\mathbf{r}}(\mathbf{r} | \mathbf{s}_i) = -\frac{N}{2} \ln(\pi N_0) - \frac{1}{N_0} \|\mathbf{r} - \mathbf{s}_i\|^2$$

$$\hat{s} = \arg \min_i \|\mathbf{r} - \mathbf{s}_i\|^2 \quad (\text{ML})$$

$$\hat{s} = \arg \min_i \left\{ \|\mathbf{r} - \mathbf{s}_i\|^2 - N_0 \ln P(\mathbf{s}_i) \right\} \quad (\text{MAP})$$

In words: choose the signal point closest to what arrived. With unequal priors, subtract a fixed handicap from each distance first — the term is positive and larger for the more likely symbol, so that symbol wins from further away.

Expanding the squared distance gives the form a receiver is actually built in. The term $\|\mathbf{r}\|^2$ is the same for every i and drops out; turning the minimum into a maximum and dividing by two leaves the **correlation metric**.

THE RECEIVER AS IT IS BUILT

$$\hat{s} = \arg \max_i \left\{ \mathbf{r} \cdot \mathbf{s}_i - \frac{E_i}{2} + \frac{N_0}{2} \ln P(\mathbf{s}_i) \right\}$$

And $\mathbf{r} \cdot \mathbf{s}_i = \int_0^T r(t) s_i(t) dt$, so the receiver can correlate against the waveforms directly and never compute coordinates. If the signals are equally likely **and** have equal energy, both corrections vanish and the rule is: take the largest correlation.

WHEN THE ENERGIES DIFFER, KEEP THE ENERGY TERM

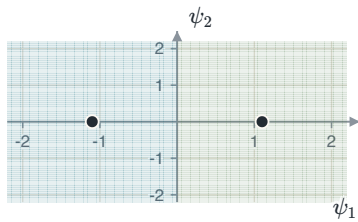
Without $-E_i/2$ a high-energy signal wins too often, because a large $\mathbf{s}_i$ makes $\mathbf{r} \cdot \mathbf{s}_i$ large whatever arrived. On-off signalling is exactly this case, and dropping the term there makes the receiver decide "one" almost always. Equal priors remove the prior term; only equal *energies* remove the energy term.

4.3 Decision regions

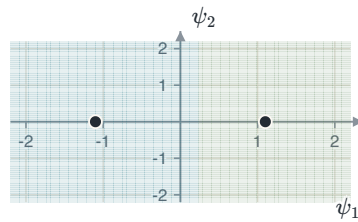
Collecting the observations that give the same answer divides the space into M regions, $R_i = \{\mathbf{r} : \|\mathbf{r} - \mathbf{s}_i\| \leq \|\mathbf{r} - \mathbf{s}_j\| \text{ for all } j\}$. Three facts describe them, and all three follow from the rule being "nearest point".

1. A boundary between two points is **perpendicular** to the line joining them.
2. With equal priors it crosses that line exactly **halfway**.
3. With unequal priors it moves, and **the region of the less likely signal shrinks**.

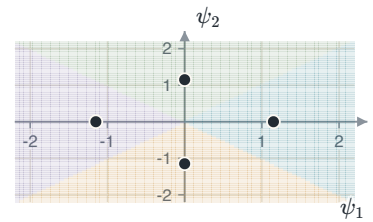
The first two are the definition of a perpendicular bisector: the set of points equidistant from two fixed points *is* that bisector. Nothing has to be calculated. Only nearest neighbours contribute boundaries, so a point in the middle of a constellation has a bounded region and a point on the outside has one that runs off to infinity — which is why outer points make fewer errors.



Two points, equal priors: the boundary is halfway.



The left symbol four times more likely: its region has grown.



Four points, four boundaries, each perpendicular to the pair it separates.

For two equally likely points a distance d apart, the boundary is $d/2$ from each. An error happens when the noise along the line joining them carries the observation across, and the noise on any one axis is $\mathcal{N}(0, N_0/2)$.

THE BINARY RESULT, FROM THE PICTURE

$$P_e = Q\left(\frac{d/2}{\sqrt{N_0/2}}\right) = Q\left(\sqrt{\frac{d^2}{2N_0}}\right)$$

$$\mu = \frac{d}{2} + \frac{N_0}{2d} \ln \frac{P(\mathbf{s}_1)}{P(\mathbf{s}_0)}$$

The second line is where the boundary sits when the priors are unequal, measured from the first point along the line. Equal priors give $\mu = d/2$.

THE RESULT THIS CHAPTER EXISTS FOR

The error probability of a binary system depends on **the distance between the two points and on nothing else**. Antipodal points are $2\sqrt{E_b}$ apart, giving $Q\left(\sqrt{2E_b/N_0}\right)$; on-off or orthogonal points are $\sqrt{2E_b}$ apart, smaller by $\sqrt{2}$ — which is the 3 dB of Chapter 2, obtained here by measuring a picture instead of integrating two densities.

4.4 The union bound

For more than two signals the exact answer is an integral of the likelihood over each decision region, in N dimensions and over a polygon with as many faces as the point has neighbours. There is no closed form. The way round it is to bound the answer instead.

Suppose $\mathbf{s}_k$ was sent, and let A_{kj} be the event that the observation is closer to $\mathbf{s}_j$ than to $\mathbf{s}_k$. An error happens exactly when at least one of those occurs, and the probability of a union is at most the sum of the probabilities. Each term is a two-point question whose answer is already known.

THE UNION BOUND AND ITS THREE USABLE FORMS

$$P(\mathbf{s}_k \rightarrow \mathbf{s}_j) = Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right)$$

$$P_e \leq \frac{1}{M} \sum_k \sum_{j \neq k} Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right)$$

$$P_e \leq (M-1)Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right), \quad P_e \approx N_{\min}Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right)$$

$N_{\min}$ is the *average* number of points at the minimum distance, and need not be an integer. The last form is the one used in practice.

One more tightening is available, and it costs nothing. To land nearer a different point the observation must leave its own region, and it leaves through a **face** — a piece of boundary shared with one particular neighbour. A point that shares no face with the region cannot be the first one reached, so the union need only run over the neighbours that bound the region. Write $\mathcal{N}(k)$ for those.

THE INTELLIGENT UNION BOUND

$$P(\text{error} \mid \mathbf{s}_k) \leq \sum_{j \in \mathcal{N}(k)} Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right)$$

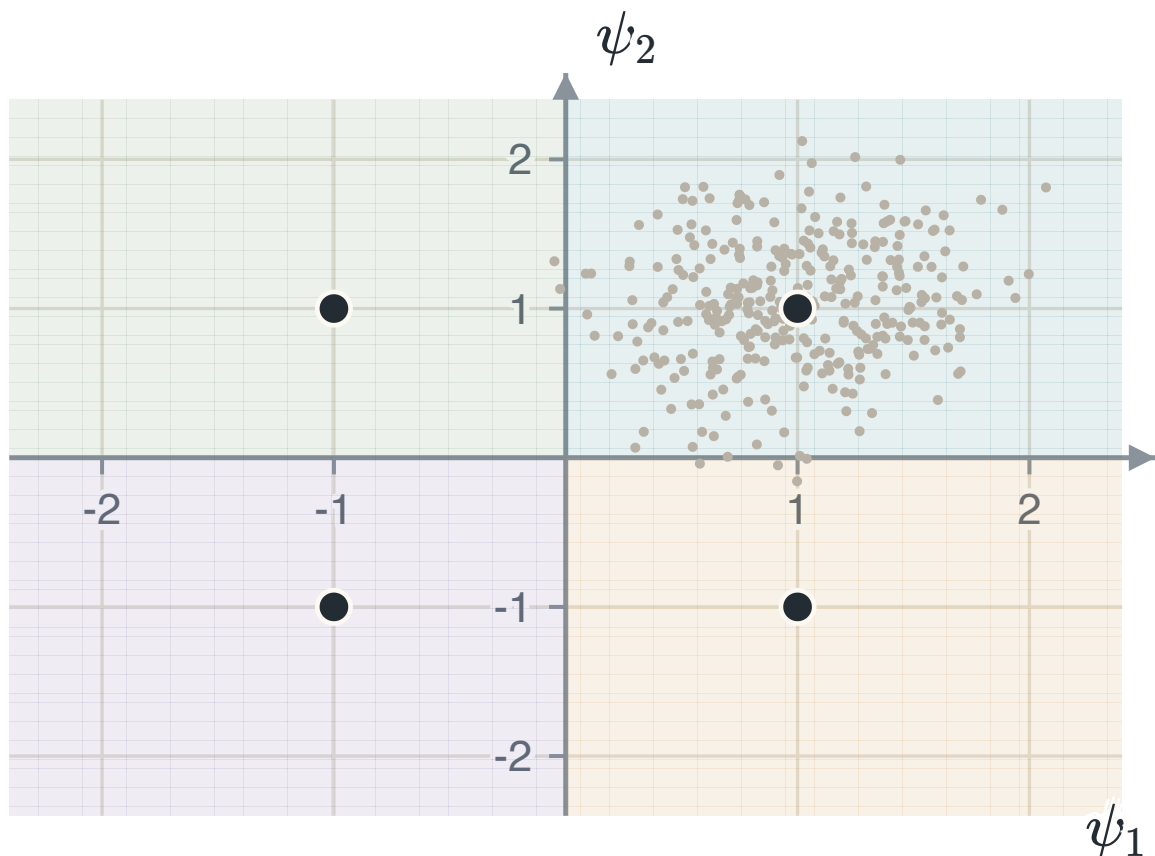
Still an upper bound, because leaving the region means crossing one of its faces. It is not the nearest-neighbour form: this one counts *faces* and bounds, that one counts *points at $d_{\min}$* and approximates. They agree whenever every face sits at $d_{\min}$, and part company when a face is shared with a point further away.

WHY IT IS AN OVER-ESTIMATE, AND WHEN THAT MATTERS

The events overlap: an observation can be closer to two other points at once, and the sum counts it twice. So the bound is always at least the truth. At low signal-to-noise ratio the overlaps are large and the bound can exceed one, which is useless; at the ratios real systems run at, every term is tiny and the bound is close enough to be quoted as the answer.

EXAMPLE 4.1 – FOUR BOUNDS ON ONE CONSTELLATION

GIVEN	Four equally likely points at $(\pm d/2, \pm d/2)$: neighbours d apart, diagonals $d\sqrt{2}$.
FIND	The symbol error probability by each of the four forms.
METHOD	Take one point, list its distances to the other three, and add one Q for each — then ask which of those three actually bound its region. Symmetry makes every point give the same answer.
SOLUTION	General: $P_e \leq 2Q\left(\sqrt{d^2/2N_0}\right) + Q\left(\sqrt{2d^2/2N_0}\right)$. Intelligent (the region has two faces, both at d): $P_e \leq 2Q\left(\sqrt{d^2/2N_0}\right)$. Nearest neighbours ($d_{\min} = d$, $N_{\min} = 2$): $P_e \approx 2Q\left(\sqrt{d^2/2N_0}\right)$. Minimum distance ($M - 1 = 3$): $P_e \leq 3Q\left(\sqrt{d^2/2N_0}\right)$.
CHECK	Put $d^2/2N_0 = 9$, so $Q(3) = 1.35 \times 10^{-3}$ and $Q(4.243) = 1.10 \times 10^{-5}$. The four answers are 2.71×10^{-3} , 2.70×10^{-3} , 2.70×10^{-3} and 4.05×10^{-3} : the diagonal term is worth 0.4%, and pretending it is a nearest neighbour costs 50%. The middle two agree here because both faces sit at $d_{\min}$, and only one of them is a bound. All four are the same Q with a different count in front, so none of them moves the curve sideways.



The constellation, its four regions, and the observations the receiver sees when the top-right point is sent. The exact error probability is the fraction of that cloud outside its own region; the bound adds three separate two-point questions and counts the overlaps twice.

4.5 Summary

RESULT	STATEMENT	ANCHOR
Observation	$\mathbf{r} = \mathbf{s}_i + \mathbf{n}$, each $n_k \sim \mathcal{N}(0, N_0/2)$, independent	PS CH8.4.1
MAP rule	minimise $\ \mathbf{r} - \mathbf{s}_i\ ^2 - N_0 \ln P(\mathbf{s}_i)$	PS CH8.4.1
ML rule	minimise $\ \mathbf{r} - \mathbf{s}_i\ ^2$: choose the nearest point	PS CH8.4.1
Receiver	maximise $\mathbf{r} \cdot \mathbf{s}_i - E_i/2 + \frac{N_0}{2} \ln P(\mathbf{s}_i)$	PS CH8.4.1
Regions	perpendicular bisectors; the less likely region shrinks	PS CH8.4.1
Binary	$P_e = Q\left(\sqrt{d^2/2N_0}\right)$	PS CH8.3.3
Union bound	$P_e \leq \frac{1}{M} \sum_k \sum_{j \neq k} Q(\cdot)$	PS CH8.4.2
Intelligent form	sum over the neighbours that give R_k a face	PS CH8.4.2
In practice	$P_e \approx N_{\min} Q\left(\sqrt{d_{\min}^2/2N_0}\right)$	PS CH8.4.2

A constellation is judged by two numbers: $d_{\min}$, which sets the exponent and therefore almost everything, and $N_{\min}$, which multiplies it and matters far less. Chapter 5 applies this to the constellations that are actually used, and asks which of them gets the largest $d_{\min}$ for a given average energy and a given number of bits per symbol.

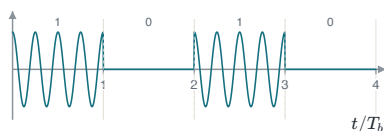
CHAPTER 5

Digital modulation methods

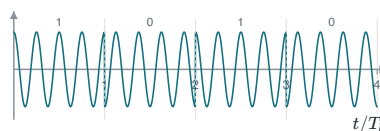
A baseband pulse cannot be sent over a radio channel, so the bits have to ride on a carrier. There are only three things about a sine wave that can be changed: its amplitude, its phase and its frequency. Every scheme in this chapter changes one of them, or two at once. Chapter 4 already gave the receiver and the error probability; all that is left is to place the points and measure the distance between them.

5.1 The three binary schemes

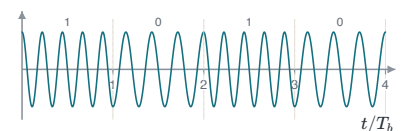
Take one bit at a time and two waveforms. Amplitude-shift keying sends the carrier for a one and nothing for a zero. Phase-shift keying sends the carrier either way and flips its sign. Frequency-shift keying sends one of two frequencies.



Amplitude-shift keying: the carrier is switched on and off.



Phase-shift keying: the carrier is always on and its sign flips.



Frequency-shift keying: one frequency for a one, another for a zero.

To compare them, put each on the axes of chapter 3 and measure. BPSK needs one basis function and its two points are $\pm\sqrt{E_b}$ on it. BFSK needs two, because the two frequencies are orthogonal, and its points

sit one on each axis. BASK needs one, and its points are 0 and $\sqrt{2E_b}$ once the average over equally likely bits is taken.

THE THREE DISTANCES AND THE THREE ERROR PROBABILITIES

$$\text{BPSK: } d_{\min}^2 = 4E_b, \quad P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

$$\text{BFSK and BASK: } d_{\min}^2 = 2E_b, \quad P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$

The two answers differ by a factor of two inside the square root, which is 3.01 dB. That gap does not depend on the noise level and cannot be closed by spending more power — it is a fact about where the points are.

WHERE THE THREE DECIBELS COME FROM

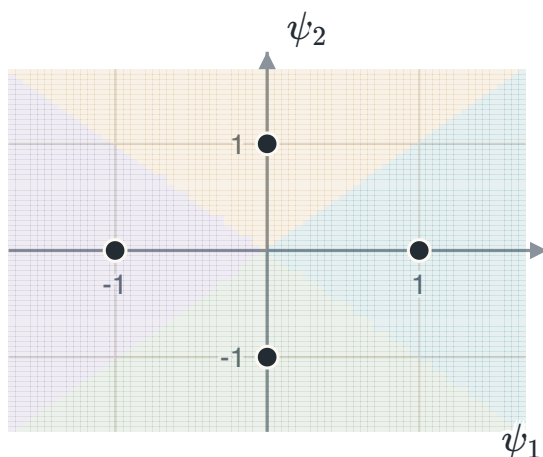
Antipodal points are $2\sqrt{E_b}$ apart. Orthogonal points are $\sqrt{2E_b}$ apart, which is smaller by $\sqrt{2}$. Squaring that $\sqrt{2}$ gives the factor of two in the Q argument, and $10 \log_{10} 2 = 3.01$ dB. Every appearance of three decibels in this chapter traces back to one $\sqrt{2}$ in a picture.

BASK AND BFSK REACH THE SAME ANSWER BY DIFFERENT ROUTES

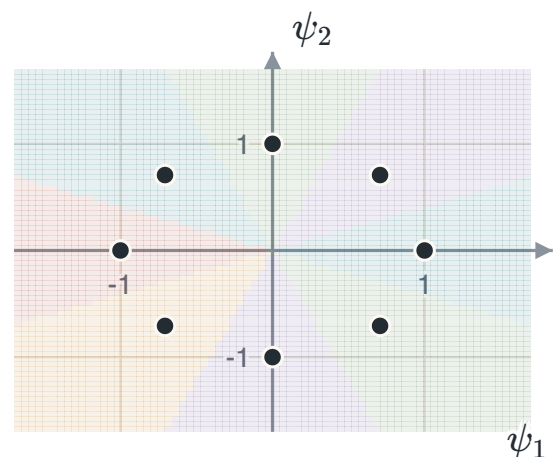
BASK keeps its points on one line but half its symbols cost nothing, so the average energy is half the peak. BFSK keeps the full energy in every symbol but separates the points by a right angle instead of a straight line. Both end at $d^2 = 2E_b$. The mathematics does not prefer either; the transmitter does, and BASK asks it to switch between zero power and full power.

5.2 M-ary phase-shift keying

Sending one bit at a time wastes the two dimensions a carrier offers. Use both, keep the energy fixed, and the constellation becomes M points spaced evenly around a circle of radius $\sqrt{E_s}$. Each symbol now carries $\log_2 M$ bits.



$M = 4$, two bits a symbol. The four regions are the four quadrants.



$M = 8$, three bits. The regions are wedges, and they are narrowing.

Two neighbouring points subtend an angle $2\pi/M$ at the centre. Dropping a perpendicular from the centre onto the line joining them cuts that angle in half and gives the distance directly.

M-PSK

$$d_{\min} = 2\sqrt{E_s} \sin\left(\frac{\pi}{M}\right), \quad N_{\min} = 2$$

$$P_e \approx 2Q\left(\sqrt{\frac{2E_s}{N_0}} \sin \frac{\pi}{M}\right)$$

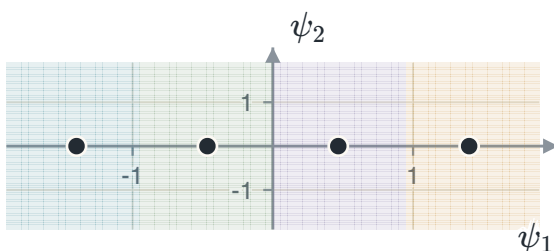
$N_{\min} = 2$ for every M above two, because a point on a circle has one neighbour clockwise and one anticlockwise. At $M = 2$ the formula gives $d_{\min} = 2\sqrt{E_s}$, which is BPSK — a general result that reproduces the case already known.

EXAMPLE 5.1 – WHAT THE STEP FROM FOUR POINTS TO EIGHT COSTS

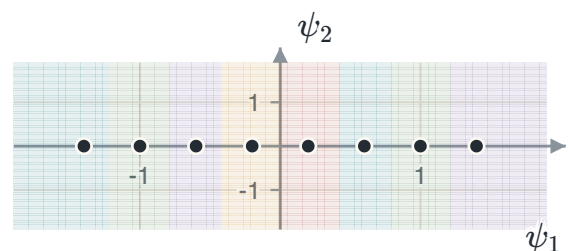
GIVEN	QPSK and 8-PSK, at whatever energy each needs to reach the same symbol error probability.
FIND	The extra energy 8-PSK needs, per symbol and per bit.
METHOD	A fixed error probability means a fixed Q argument, so $(E_s/N_0) \sin^2(\pi/M)$ must be held constant. Take the ratio.
SOLUTION	$\frac{\sin^2(\pi/4)}{\sin^2(\pi/8)} = \frac{0.5}{0.1464} = 3.414$, which is 5.33 dB per symbol. 8-PSK carries three bits where QPSK carries two, so per bit the figure is $3.414 \times \frac{2}{3} = 2.276$, or 3.57 dB.
CHECK	Both numbers are positive, as they must be: at the same radius, eight points are closer together than four. The per-bit figure is the smaller of the two, because the extra bit pays part of its own way.

5.3 M-ary amplitude-shift keying

Change the amplitude among M levels instead. One basis function carries them all, so the constellation is M points on a line at $\pm A, \pm 3A, \dots, \pm(M-1)A$, and neighbouring points are $2A$ apart.



$M = 4$ on one axis. The two end regions run off to infinity.



$M = 8$: the same line, twice as crowded.

$$E_{s,\text{avg}} = \frac{A^2(M^2 - 1)}{3} \implies d_{\min}^2 = \frac{12E_{s,\text{avg}}}{M^2 - 1}$$

$$N_{\min} = \frac{2(M - 1)}{M}, \quad P_e \approx N_{\min} Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right)$$

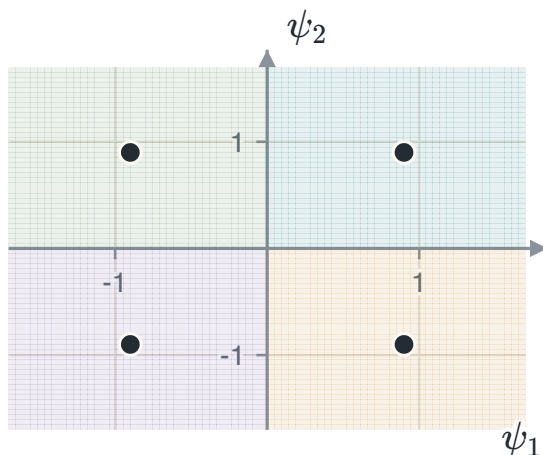
The M^2 in the denominator is the trouble: doubling M divides the squared distance by roughly four, which is about 6 dB, and it does so at every doubling.

WHY $N_{\min}$ IS NOT A WHOLE NUMBER

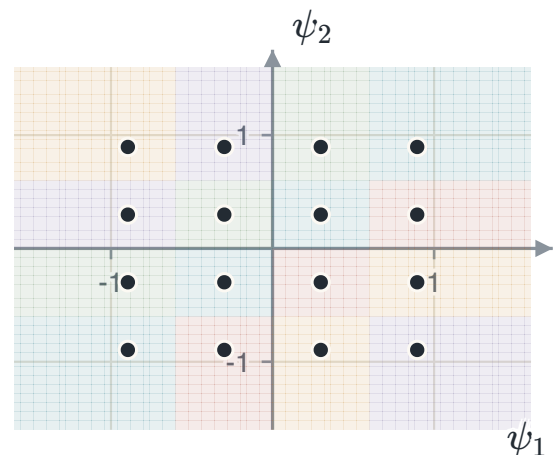
The two points at the ends of the line have one neighbour each; the $M - 2$ points between them have two. Averaging over all M equally likely symbols gives $2(M - 1)/M - 1.5$ at $M = 4$, 1.75 at $M = 8$. It is an average over points that differ in how exposed they are, and getting a whole number means the end points were counted as though they had neighbours on both sides.

5.4 Quadrature amplitude modulation

Run one amplitude-modulated signal on the cosine and an independent one on the sine. The result is two $\sqrt{M}$ -level constellations at right angles, so M points on a square grid — the lecture material states this as M -QAM being two-dimensional M -ASK.



4-QAM: two levels a side. These are the four points of QPSK — at $M = 4$ the two schemes are the same scheme.



16-QAM: four levels a side, four bits a symbol. The corners have two neighbours, the edges three, the four in the middle four.

SQUARE M-QAM

$$E_{s,\text{avg}} = \frac{(M-1)d^2}{6} \implies d_{\min}^2 = \frac{6E_{s,\text{avg}}}{M-1}$$

$$N_{\min} = \frac{4(2) + 8(3) + 4(4)}{16} = 3 \quad \text{for 16-QAM}$$

Compare the denominators: $M-1$ for QAM against M^2-1 for PAM. Spreading the same number of points over two dimensions instead of one turns a square into a linear factor, and that is the whole of the difference.

EXAMPLE 5.2 – SIXTEEN POINTS ON A GRID AGAINST SIXTEEN ON A LINE

GIVEN	16-QAM and 16-PAM at the same average symbol energy. Both carry four bits a symbol.
FIND	The advantage of QAM in decibels.
METHOD	One formula each, then the ratio of the squared distances.
SOLUTION	QAM: $d_{\min}^2 = 6E_s/15 = 0.400E_s$. PAM: $d_{\min}^2 = 12E_s/255 = 0.0471E_s$. The ratio is 8.5, and $10 \log_{10} 8.5 = 9.29$ dB.
CHECK	Same number of bits, same average power, and the only difference is where the points were put. Nine decibels is an enormous return for using a second dimension that was there all along, and it is the reason QAM is what modern systems use.

5.5 M-ary frequency-shift keying

Use M frequencies, chosen so that the waveforms are orthogonal. Orthogonal signals need one basis function each, so the constellation lives in M dimensions with one point on each axis at distance $\sqrt{E_s}$ from the origin.

M-FSK

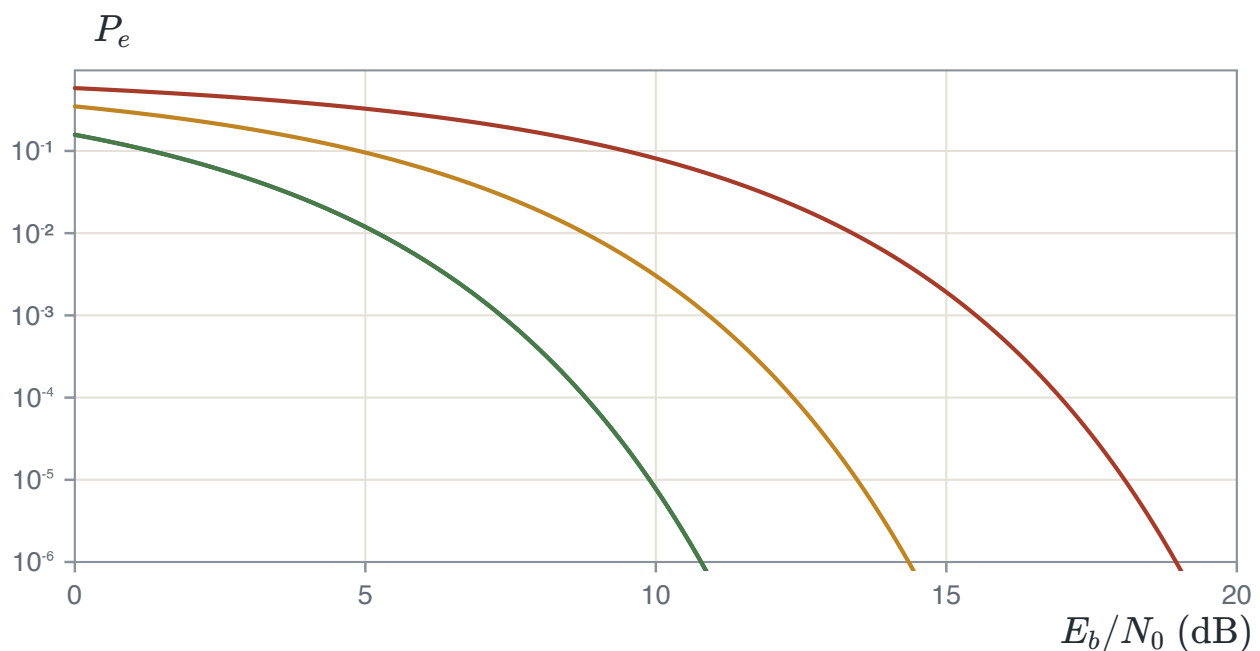
$$d_{jk} = \sqrt{2E_s} \quad \text{for every pair,} \quad N_{\min} = M-1$$

$$P_e \approx (M-1) Q\left(\sqrt{\frac{E_s}{N_0}}\right)$$

The distance does not depend on M at all. Adding waveforms costs nothing in distance — but each new waveform needs its own frequency slot, so the price is paid in bandwidth instead.

THE TRADE, STATED PLAINLY

PSK and QAM hold the bandwidth fixed as M grows and pay in energy, because the points crowd together. FSK holds the distance fixed and pays in bandwidth. A deep-space link, with bandwidth to spare and no power, chooses FSK; a mobile phone, with the opposite problem, chooses QAM. Neither is better in the abstract — it depends entirely on which resource is scarce.



Symbol error probability against energy per bit for four sizes of PSK: $M = 2, 4, 8, 16$ from left to right. $M = 2$ and $M = 4$ lie almost on top of each other, which is why QPSK is everywhere — it carries twice the bits of BPSK for the same energy per bit.

5.6 Summary

SCHEME	$d_{\min}^2$	$N_{\min}$	ANCHOR
BPSK	$4E_b$	1	PS CH8.6.1
BFSK, BASK	$2E_b$	1	PS CH9.5
M -PSK	$4E_s \sin^2(\pi/M)$	2	PS CH8.6.3
M -PAM	$12E_s/(M^2 - 1)$	$2(M - 1)/M$	PS CH8.5.3
M -QAM	$6E_s/(M - 1)$	3 at $M = 16$	PS CH8.7.1
M -FSK	$2E_s$	$M - 1$	PS CH9.5

Every row was obtained the same way: write the waveforms down, find the basis functions they need, place the points, measure the shortest distance, count how many pairs sit at it. The error probability then comes from chapter 4 without any further thought about waveforms. Convert once with $E_s = (\log_2 M)E_b$ when a question is stated per bit, and never convert twice.

What none of this says is how many bits a channel can carry at all. Every scheme here trades energy against bandwidth, and it is reasonable to ask whether there is a limit to the trade. Chapter 6 answers that, and the answer does not depend on which modulation is used.

CHAPTER 6

An introduction to information theory

Every chapter so far has taken the bits as given and worked on getting them across a channel. This one asks where they came from and how few of them a message really needs. The answer is one number, computed

from the source alone, and it is both a measure of how much information the source produces and a limit on how few bits will carry it.

6.1 The information in one symbol

A **discrete memoryless source** emits one of K symbols each time, with fixed probabilities $p_1, \dots, p_K$, chosen independently of what came before. How much is learnt when one arrives? If it was certain, nothing. If it was almost impossible, a great deal.

SELF-INFORMATION

$$I(s_k) = \log_a \frac{1}{p_k} = -\log_a p_k$$

The base sets the unit: $a = 2$ gives **bits**, $a = e$ gives **nats**, $a = 10$ gives **Hartleys**. This course uses base two throughout.

THREE PROPERTIES, AND WHAT FORCES THE LOGARITHM

$I(s_k) \geq 0$, because a probability is at most one and its logarithm at most zero.

$I(s_k) \geq I(s_j)$ **when** $p_k \leq p_j$: rarer means more informative.

$I(s_k s_j) = I(s_k) + I(s_j)$ **for independent symbols**, because their probabilities multiply and the logarithm turns a product into a sum. That third property is the one that rules out every function except a logarithm.

ONE HALVING IS ONE BIT

$-\log_2 \frac{1}{2} = 1$, so every halving of a probability adds exactly one bit. A symbol of probability $1/1000$ is about ten halvings away from certainty, so it carries about 10 bits. Most estimates in this chapter can be done that way, without a calculator.

6.2 Entropy

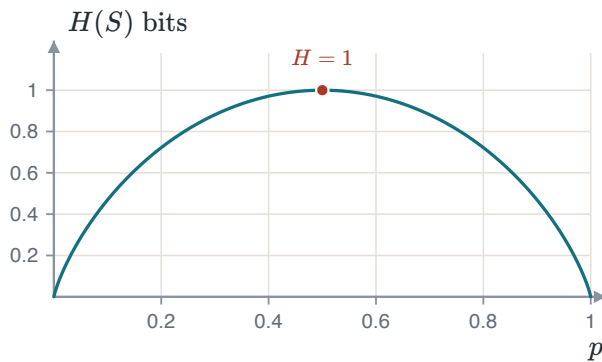
Self-information describes one symbol. Average it over the alphabet, weighting each symbol by how often it happens, and the result describes the source.

ENTROPY

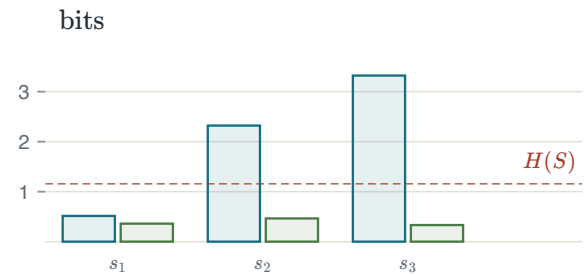
$$H(S) = E[I(s_k)] = - \sum_{k=1}^K p_k \log_2 p_k \quad \text{bits a symbol}$$

$$0 \leq H(S) \leq \log_2 K$$

Zero when one symbol has probability one — nothing is ever in doubt. At the top, $\log_2 K$, when all K are equally likely: putting $p_k = 1/K$ into the sum gives $\sum \frac{1}{K} \log_2 K = \log_2 K$.



The entropy of a binary source. One bit at $p = \frac{1}{2}$, nothing at either end, and flat near the top — a slightly unfair coin is almost as informative as a fair one.



The standard example. The left bar of each pair is what the symbol carries; the right bar is that weighted by how often it happens. The three right bars add to $H(S)$.

EXAMPLE 6.1 – THE ENTROPY OF A THREE-SYMBOL SOURCE

GIVEN $S = \{s_1, s_2, s_3\}$ with probabilities 0.7, 0.2, 0.1.

FIND The entropy.

SOLUTION $H(S) = -0.7 \log_2 0.7 - 0.2 \log_2 0.2 - 0.1 \log_2 0.1 = 0.3602 + 0.4644 + 0.3322 = 1.1568$ bits a symbol.

CHECK $\log_2 3 = 1.585$, and 1.1568 is below it, as it must be for a source that is not uniform. Numbering the three symbols would cost 2 bits each, so plain numbering wastes almost 0.85 bits every symbol. Recovering that is the whole of source coding.

Two numbers get confused here and should not be. $\log_2 K$ is the most a source of this size *could* carry. $\lceil \log_2 K \rceil$ is what a fixed-length code *costs*. The first gap is the source being uneven; the second is rounding.

6.3 Extended sources

Nothing forces a coder to work one symbol at a time. Group them in n s and treat each block as one symbol of a new alphabet. That is the **n -th extension**, written S^n , and it has K^n symbols.

EXTENSION

$$H(S^n) = n H(S)$$

The source is memoryless, so the symbols in a block are independent and their information adds. This is the third property of self-information, applied n times.

EXAMPLE 6.2 – THE SAME SOURCE, TWO AT A TIME

GIVEN	The source of Example 6.1, extended by two.
FIND	$H(S^2)$.
METHOD	S^2 has $3^2 = 9$ symbols with probabilities 0.49, 0.14, 0.07, 0.14, 0.04, 0.02, 0.07, 0.02, 0.01.
SOLUTION	Summing $-p \log_2 p$ over all nine gives 2.3136 bits a block.
CHECK	$2 \times 1.1568 = 2.3136$. The long way and the short way agree, and per symbol nothing has changed — 2.3136 over two symbols is 1.1568 each. What changes is how much of a whole bit gets wasted in rounding, and section 6.6 turns that into a bound.

ONLY BECAUSE IT IS MEMORYLESS

If the source had memory — as English does, where q is followed by u — the block probabilities would not be products, and $H(S^n)$ would be *less* than $nH(S)$. That difference is exactly the redundancy real compressors live on, and it is outside this course.

6.4 What a code costs

A source encoder turns each symbol into a string of bits, its **codeword**. The codewords need not all be the same length, and the good idea of this chapter is that they should not be: short ones for common symbols, long ones for rare.

AVERAGE LENGTH AND EFFICIENCY

$$\bar{L} = \sum_{k=1}^K p_k l_k, \quad \eta = \frac{L_{\min}}{\bar{L}} = \frac{H(S)}{\bar{L}} \leq 1$$

The **source-coding theorem** says $\bar{L} \geq H(S)$ for any code from which the symbols can be recovered exactly. So $L_{\min} = H(S)$: the entropy is not merely a measure of information, it is a limit on how few bits will carry it.

SOURCE CODING IS NOT CHANNEL CODING

Source coding removes bits the message does not need. Channel coding adds bits back so that errors can be found and fixed. They pull in opposite directions, they are done in that order, and this course covers only the first.

Written English gives a concrete case. Its entropy is estimated at about 1.3 bits a letter, while a typical letter-by-letter variable-length code reaches $\bar{L} = 4.22$, so $\eta = 0.31$. Morse code is the same idea by hand: one dot for e , four symbols for z .

6.5 Codes that can be read back

Short codewords are worth nothing if the receiver cannot tell where one ends. Two conditions matter, and they are not the same condition. A code is **uniquely decodable** if every string of bits it can produce comes from exactly one string of symbols. It is a **prefix code** if no codeword is the beginning of any other —

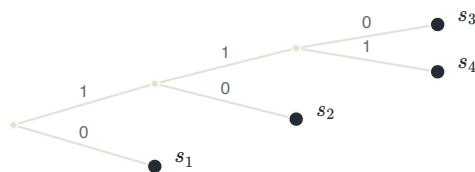
which is stronger, and which lets the decoder name a symbol the moment its last bit arrives. A prefix code is also called **instantaneous**.

SYMBOL	CODE I	CODE II	CODE III
s_1	0	0	0
s_2	1	10	01
s_3	00	110	011
s_4	11	111	0111

Code I is not a prefix code and is not uniquely decodable either: the string 00 is both s_3 and s_1s_1 . Code II is a prefix code. Code III is not a prefix code, since 0 begins all three of the others.

CODE III IS THE INTERESTING ONE

It *can* be read back: every codeword begins with the only 0 and is then named by how many 1s follow, so any concatenation parses one way. What it cannot do is decode on the fly — after reading 011 the decoder still does not know whether it has s_3 or the start of s_4 , and it must wait for the next 0. Uniquely decodable, but not instantaneous. The distinction is the whole point of the example.



Code II. Every symbol is at a leaf, so no path to one passes through another. That is what the prefix property looks like.



Code III. Every symbol sits on one path, each hanging off the node before it. Nothing is at a leaf but the last, and that is why the decoder must wait.

6.6 The Kraft inequality, and how close a code can get

Before choosing any codewords, ask whether the *lengths* can work. Think of the whole tree as one unit: a codeword of length l claims a branch and everything below it, a fraction 2^{-l} of the tree, and the claims cannot add to more than the tree there is.

KRAFT INEQUALITY

$$\sum_{k=1}^K 2^{-l_k} \leq 1$$

For the three codes above: 1.5, 1 and 0.9375. Code I fails, so no prefix code has its lengths. Code II uses the tree exactly. Code III passes with a sixteenth to spare — and is still not a prefix code, because the inequality says a prefix code with those *lengths* exists, not that these particular codewords are one. It is **necessary but not sufficient**.

HOW CLOSE A PREFIX CODE GETS

$$H(S) \leq \bar{L} < H(S) + 1$$

$$H(S) \leq \frac{L_n}{n} < H(S) + \frac{1}{n} \implies \lim_{n \rightarrow \infty} \frac{L_n}{n} = H(S)$$

The ideal length for symbol k is $-\log_2 p_k$, almost never a whole number, so each length is rounded up and the rounding costs under one bit. Applying the same bound to the n -th extension spreads that one bit over n symbols.

WHEN THE BOUND IS TIGHT

If every probability is a power of two — a **dyadic** distribution, $p_k = 2^{-l_k}$ — nothing needs rounding. Then $\bar{L} = \sum l_k 2^{-l_k}$ and $H(S) = -\sum 2^{-l_k} \log_2 2^{-l_k} = \sum l_k 2^{-l_k}$: the same sum. So $\bar{L} = H(S)$ exactly and the code is perfect.

WHAT BLOCKING COSTS

The bound improves as $1/n$ and the alphabet grows as K^n . Reaching within 0.05 bits a symbol needs $n = 20$, and for a three-symbol source that is $3^{20} = 3.5 \times 10^9$ codewords. The theory says the entropy is reachable; the arithmetic says not this way, and both are true.

6.7 Huffman coding

The bound says a good code exists. Huffman coding builds one, and it builds the best one: no prefix code on single symbols has a smaller average length.

THE ALGORITHM

1. List the symbols in order of decreasing probability.
 2. Take the two least likely, label one 0 and the other 1, and merge them into one symbol whose probability is their sum. Re-sort.
 3. Repeat until two symbols remain, and label those 0 and 1.
- Each codeword is the labels along the path back to its symbol.

EXAMPLE 6.3 — THE STANDARD FIVE-SYMBOL SOURCE

GIVEN $p = 0.4, 0.2, 0.2, 0.1, 0.1$.

MERGING $0.1 + 0.1 = 0.2$; then the two smallest are 0.2 and 0.2, giving 0.4; then 0.2 and 0.4 give 0.6; then 0.4 and 0.6 finish it.

SOLUTION Lengths 2, 2, 2, 3, 3, so $\bar{L} = 0.4(2) + 0.2(2) + 0.2(2) + 0.1(3) + 0.1(3) = 2.2$ bits a symbol.

EFFICIENCY $H(S) = 2.1219$ bits, so $\eta = 2.1219/2.2 = 0.9645$ — the code spends 3.68% more than the source strictly needs.

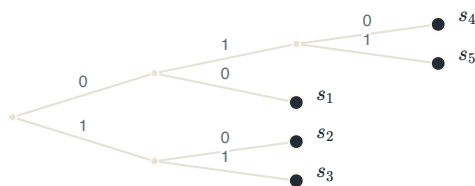
CHECK $2.1219 \leq 2.2 < 3.1219$: the two-sided bound holds. A fixed-length code would have cost $\lceil \log_2 5 \rceil = 3$ bits, so Huffman saved 0.8 bits a symbol.

Two choices in the algorithm are free: which of a merged pair gets 0, and where a merged symbol goes when it ties with one already in the list. Different choices give different codes with the *same* average length — they must, because the minimum is unique even when the code is not. What separates them is how far the lengths spread about that average.

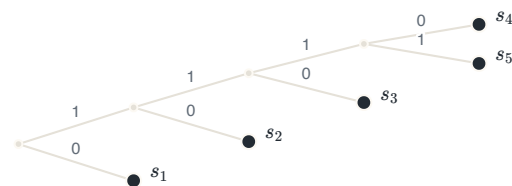
VARIANCE OF THE CODEWORD LENGTH

$$\sigma^2 = \sum_{k=1}^K p_k (l_k - \bar{L})^2$$

For the example: placing the merged symbol high gives lengths 2, 2, 2, 3, 3 and $\sigma^2 = 0.16$; placing it low gives 1, 2, 3, 4, 4, the same $\bar{L} = 2.2$, and $\sigma^2 = 1.36$.



Merged symbol placed high. Lengths 2, 2, 2, 3, 3, average 2.2, variance 0.16. The tree is nearly balanced.



Merged symbol placed low. Lengths 1, 2, 3, 4, 4, the same average 2.2, variance 1.36. The tree is a ladder.

WHY THE SMALLER VARIANCE IS WANTED

The encoder produces bits at a varying rate and the channel takes them at a fixed one, so a buffer sits between them. A code with wildly different lengths makes that buffer fill and empty unpredictably; a code with steady lengths does not. Both cost 2.2 bits on average, and only one is comfortable to build. **The rule:** on a tie, move the merged symbol as high as possible in the list.

6.8 Lempel–Ziv coding

Huffman coding has one practical weakness, and it is not its length. It needs the probabilities *before* it can build the tree, and for a stream arriving over a wire nobody knows them in advance. A **universal** code is one that reaches the entropy without being told the source, and the Lempel–Ziv algorithm is the one in use everywhere.

THE ALGORITHM

Read the stream once. Each time, take the **shortest run of bits not seen before** and store it in a dictionary. Because it is the shortest new one, everything but its last bit is already stored — so it is sent as a **pointer** to that earlier entry followed by the one new bit, the **innovation**. The decoder builds the same dictionary from the same blocks in the same order, so the dictionary is never transmitted.

EXAMPLE 6.4 – PARSING AND SENDING A STREAM

GIVEN	The stream 0 1 00 011 1 . . . , with 0 and 1 held in the dictionary at positions 1 and 2.
METHOD	Parse left to right into pieces not seen before, then send each piece as (position of its start, new bit).
SOLUTION	00 takes position 3: its start 0 is at position 1 and its new bit is 0, so with four-bit blocks it goes as 001 0. Next 011 takes position 4: its start 01 is at position 4... so it is sent as 100 1.
DECODING	Receive 1101. The last bit is the innovation, 1. The first three, 110, are 6 in binary, so the piece is entry 6 followed by 1 — and entry 6 is already in the decoder's own dictionary.
CHECK	Seven pieces at four bits each cost 28 bits where the raw stream was 18. The method <i>loses</i> on a short stream, and that is not a fault: the dictionary has to be paid for before it can pay back. In practice the block is 12 bits, giving $2^{12} = 4096$ entries, and a long file compresses to roughly two thirds of its size.

OPTIMAL AGAINST UNIVERSAL

Huffman is optimal but not universal: no prefix code on single symbols beats it, and it cannot start until the probabilities are known. Lempel–Ziv is universal but reaches the limit only in the long run. That trade is the whole difference between them, and it is why the second is what compresses an archive.

6.9 The discrete memoryless channel

The chapter so far has been about the source alone. The rest is about what a channel does to it, and the first job is to write a channel down in a form that can be calculated with. A **discrete memoryless channel** takes one symbol from a finite input alphabet and produces one from a finite output alphabet. **Discrete** because both alphabets are finite; **memoryless** because the output depends only on the input sent at that moment.

THE CHANNEL, AND WHAT THE TRANSMITTER ADDS

$$p(y_k | x_j) = P(Y = y_k | X = x_j), \quad \sum_{k=0}^{K-1} p(y_k | x_j) = 1 \text{ for every } j$$
$$p(x_j, y_k) = p(y_k | x_j) p(x_j), \quad p(y_k) = \sum_{j=0}^{J-1} p(y_k | x_j) p(x_j)$$

The transition probabilities collected into a matrix, one row per input, are the **channel matrix**. Every *row* sums to one, because something must come out when a symbol goes in; the columns do not, and expecting them to is the usual first mistake. The **input distribution** $p(x_j)$ is not part of the channel — it belongs to the transmitter, and it is chosen.

THE BINARY SYMMETRIC CHANNEL

Two inputs, two outputs, and one number: the probability p that a bit is flipped, called the **crossover probability**. Its matrix is $\begin{bmatrix} 1-p & p \\ p & 1-p \end{bmatrix}$. It is not an abstraction invented for this chapter — the binary receiver of chapter 2, seen from outside, *is* a binary symmetric channel with $p = Q\left(\sqrt{2E_b/N_0}\right)$. With equally likely inputs the output is equally likely whatever p is, even at $p = \frac{1}{2}$ where the channel is destroying everything, so a balanced output is no evidence that anything survived.

6.10 Mutual information

Before anything arrives, the uncertainty about the input is $H(X)$. Then one output symbol arrives, and it rarely settles the question but it does change the odds. What remains is an entropy like any other, averaged over which output actually came.

CONDITIONAL ENTROPY, AND THE INFORMATION THAT GOT THROUGH

$$H(X | Y) = \sum_k H(X | Y = y_k) p(y_k) = - \sum_k \sum_j p(x_j, y_k) \log_2 p(x_j | y_k)$$

$$I(X; Y) = H(X) - H(X | Y) = H(Y) - H(Y | X)$$

$$H(X, Y) = H(X) + H(Y) - I(X; Y)$$

Read the middle line in words: **how much doubt the arrival removed**, in bits, per use of the channel. The second form is the one used, because $H(Y | X)$ comes straight off the channel matrix while $H(X | Y)$ needs Bayes' rule first.

THREE PROPERTIES

Symmetry. $I(X; Y) = I(Y; X)$. Bayes' rule gives $p(x_j | y_k)/p(x_j) = p(y_k | x_j)/p(y_k)$, and substituting turns one sum into the other. What the output says about the input equals what the input says about the output — which is why the quantity is called *mutual*.

Non-negativity. $I(X; Y) \geq 0$, with equality exactly when X and Y are independent. An observation never leaves you knowing less than before.

Joint entropy. Adding $H(X)$ and $H(Y)$ counts the shared part twice, so it is subtracted once. That is the whole content of the third line.

EXAMPLE 6.5 – THE BINARY SYMMETRIC CHANNEL AT $p = 0.1$

GIVEN	A BSC with $p = 0.1$ and equally likely inputs.
FIND	$H(X)$, $H(Y X)$, $H(Y)$ and $I(X; Y)$.
METHOD	$H(Y X)$ comes off the matrix: whichever symbol goes in, the output distribution is $(0.9, 0.1)$ in some order. $H(Y)$ needs the output distribution first.
SOLUTION	$H(X) = 1$ bit. $H(Y X) = H(0.1) = -0.1 \log_2 0.1 - 0.9 \log_2 0.9 = 0.4690$ bits. The output is equally likely, so $H(Y) = 1$ bit, and $I(X; Y) = 1 - 0.4690 = 0.5310$ bits per use.
CHECK	One bit was offered and just over half a bit arrived. The missing 0.4690 bits were not delayed or corrupted — they were destroyed, and no receiver however clever recovers them. The joint entropy is $1 + 1 - 0.5310 = 1.4690$ bits, and at $p = 0.25$ the shared part falls to 0.1887 while the total rises to 1.8113: a noisier channel does not destroy uncertainty, it moves it out of the shared part.

THE TWO CONDITIONAL ENTROPIES ARE NOT THE SAME THING

$H(Y | X)$ is the uncertainty about the *output* given the input, and for the BSC it is $H(p)$ whatever the transmitter does — a property of the channel alone. $H(X | Y)$ is the uncertainty about the *input* given the output, and it depends on the input distribution too. $I(X; Y) = I(Y; X)$ is a statement about one number; it does not make the two ends interchangeable.

6.11 Channel capacity

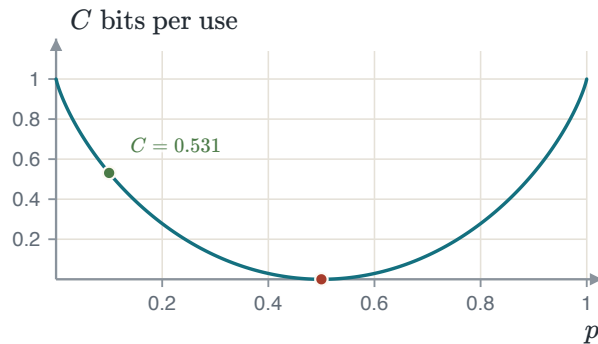
Mutual information measures one transmitter on one channel; change how often each symbol is sent and the number moves. The channel is fixed and the input distribution is the engineer's to choose, so take the best choice. What comes out then depends on the channel matrix alone, because the one free thing has been optimised away: **capacity is a property of the channel, and mutual information is not.**

CAPACITY, AND THE CAPACITY OF THE BSC

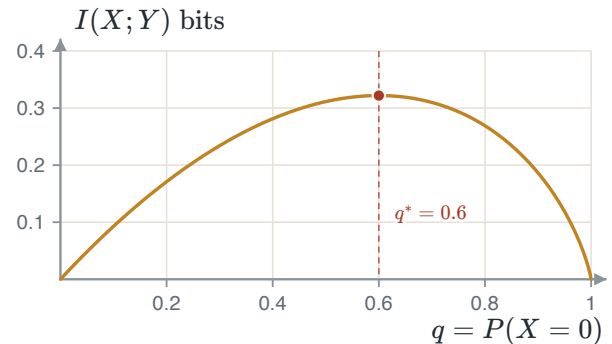
$$C = \max_{\{p(x_j)\}} I(X; Y) \quad \text{bits per channel use}$$

$$C_{\text{BSC}} = 1 - H(p), \quad H(p) = -p \log_2 p - (1 - p) \log_2 (1 - p)$$

For the BSC, $H(Y | X) = H(p)$ whatever the transmitter does, and $H(Y) \leq 1$ bit with equality when the output is equally likely — which equally likely inputs deliver. So the maximum sits at $p(x_0) = \frac{1}{2}$ and the two pieces subtract. The unit is bits per *use* of the channel, not bits per second; multiply by the number of uses a second to reach a rate.



The capacity of the binary symmetric channel. Worth a full bit at both ends — at $p = 1$ every bit is flipped, which the receiver simply undoes — and zero at $p = \frac{1}{2}$, the **useless channel**, where the output is independent of the input.



The Z-channel of Example 6.6. Its mutual information peaks at $q = 0.6$, not at a half, which is what an asymmetric channel does: the transmitter should favour the symbol that survives.

The curve is flat near $p = 0$: a channel with one error in a thousand has capacity 0.9886 bits, so the first errors cost almost nothing. It then falls away steeply — at $p = 0.11$ the capacity is already 0.500. Halving the error probability of an already-good channel buys very little, and the effort belongs where the curve is steep.

EXAMPLE 6.6 – THE CAPACITY OF THE Z-CHANNEL

- GIVEN** A 0 always arrives as a 0; a 1 arrives correctly half the time and as a 0 otherwise. Write $P(X = 0) = q$.
- FIND** The capacity, and the input distribution that achieves it.
- METHOD** Nothing is symmetric, so the equally likely input is no longer the right guess. Form $I(X; Y)$ as a function of q and maximise it.
- SOLUTION** $P(Y = 0) = q + \frac{1}{2}(1 - q) = \frac{1}{2}(1 + q)$, and $H(Y | X) = q(0) + (1 - q)(1) = 1 - q$ because a transmitted 0 leaves nothing in doubt. Subtracting gives $I(X; Y) = q - \frac{1}{2}(1 + q) \log_2(1 + q) - \frac{1}{2}(1 - q) \log_2(1 - q)$. Setting the derivative to zero collapses the logarithmic terms to $\frac{1}{2} \log_2 \frac{1-q}{1+q} = -1$, so $\frac{1-q}{1+q} = \frac{1}{4}$ and $q^* = 0.6$.
- CHECK** $C = 0.3219$ bits per channel use, and $0.3219 = \log_2 \frac{5}{4}$ exactly — a check on the arithmetic that costs nothing. The optimum is not the balanced input: the transmitter should send the reliable symbol three times in five. A transmitter that guesses $q = 0.5$ still gets 0.3113 bits, losing about 3%.

THE CHANNEL CODING THEOREM

If the transmission rate satisfies $R_b < C$, there is a coding scheme whose probability of error is as small as required. If $R_b > C$, there is not — no scheme, however long or however clever. **An error-free data rate can never exceed the capacity.** The theorem says a code exists; it does not say what the code is, how long its blocks must be, or what the decoder costs. Finding codes that come close and can also be decoded took the fifty years after it was proved.

6.12 The bandlimited channel

Everything above counted symbols. A real channel is given instead as a bandwidth in hertz, an average transmitted power and a noise density. Its capacity is one of the most quoted results in engineering.

THE INFORMATION CAPACITY LAW

$$C = B \log_2 \left(1 + \frac{P}{N_0 B} \right) \quad \text{bits per second}$$

$$\frac{C}{B} = \log_2 \left(1 + \frac{E_b C}{N_0 B} \right) \implies \frac{E_b}{N_0} = \frac{2^{C/B} - 1}{C/B}$$

B is the bandwidth, P the average transmitted power and $N_0 B$ the noise power in the band. Writing $P = E_b C$ — power is energy per bit times bits per second — and dividing by B gives the second line, in which C/B is the **bandwidth efficiency** in bits per second per hertz.

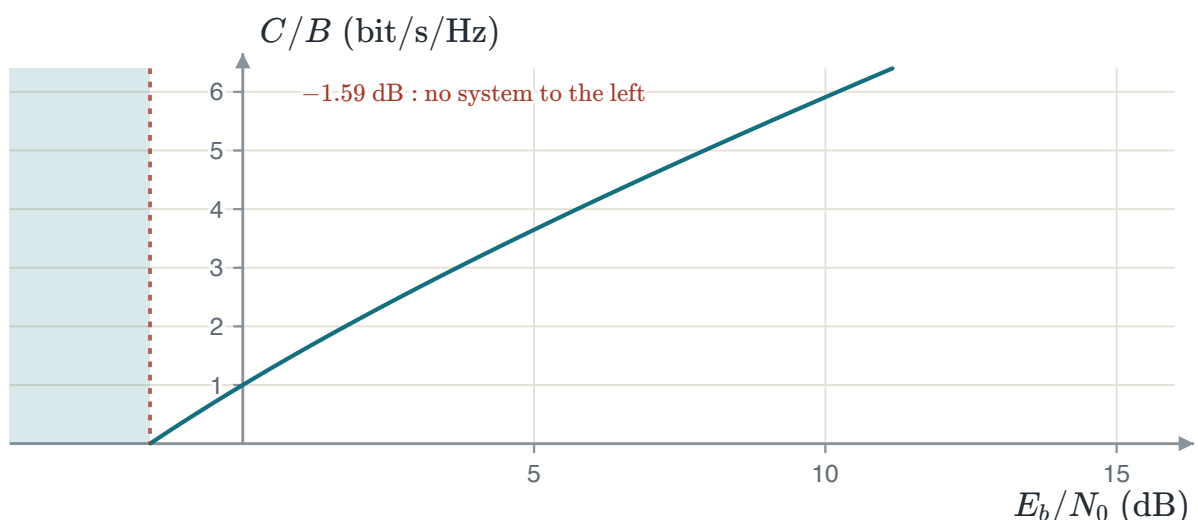
READ THE TWO KNOBS

C grows **linearly** with bandwidth and only **logarithmically** with power. Doubling the bandwidth at fixed noise density roughly doubles the rate; doubling the power adds one bit per second per hertz at best and much less when the ratio is already large. Bandwidth is the better buy, and it is the one that is scarce and regulated. At $C/B = 1$ the requirement is $E_b/N_0 = 1$, which is 0 dB; at $C/B = 2$ it is 1.5, which is 1.76 dB.

THE SHANNON LIMIT

$$\lim_{x \rightarrow 0} \frac{2^x - 1}{x} = \ln 2, \quad \left. \frac{E_b}{N_0} \right|_{\min} = \ln 2 = 0.693 = -1.59 \text{ dB}$$

Spend bandwidth freely — let $C/B \rightarrow 0$ — and the energy per bit a system needs falls, but not to nothing. No system of any kind communicates reliably below -1.59 dB: not a better code, not a better modulation, not a better receiver.



Bandwidth efficiency against energy per bit. Every working system sits below the curve and to the right of the line; the region to the left of -1.59 dB has never been occupied and never will be. Climbing from one bit per hertz to six costs about 14 dB, and the climb steepens the whole way.

A FLOOR THAT IS APPROACHED AND NEVER TOUCHED

Reaching the limit needs infinite bandwidth, and the rate per hertz goes to zero on the way. Coherent binary PSK needs about 9.6 dB for an error probability of 10^{-5} , so it sits some 11 dB above the floor. Closing that gap is what channel coding was invented for; this course stops at the uncoded schemes, which is where the gap is widest and easiest to see.

6.13 Summary

RESULT	STATEMENT	ANCHOR
Self-information	$I(s_k) = -\log_2 p_k$	PS CH12.1.1
Entropy	$H(S) = -\sum_k p_k \log_2 p_k$	PS CH12.1.1
Bounds	$0 \leq H(S) \leq \log_2 K$	PS CH12.1.1
Extension	$H(S^n) = nH(S)$	PS CH12.1.1
Average length	$\bar{L} = \sum_k p_k l_k, \eta = H(S)/\bar{L}$	PS CH12.2
Source-coding theorem	$\bar{L} \geq H(S)$, so $L_{\min} = H(S)$	PS CH12.2
Kraft	$\sum_k 2^{-l_k} \leq 1$: necessary, not sufficient	PS CH12.3
Prefix bound	$H(S) \leq \bar{L} < H(S) + 1$	PS CH12.2
Huffman	optimal, not unique; high placement gives least variance	PS CH12.3.1

Chapter 1 turned a waveform into bits. Chapters 2 to 5 got those bits across a channel and counted the errors. This chapter asked how few bits there needed to be in the first place, and answered with one number that the source itself decides. Everything between the two is engineering; the entropy is not.

APPENDIX A

Summary of formulas

Everything the course establishes, in the order it establishes it. Nothing here is derived — where a result needs an argument, the argument is in the chapter named beside it.

A.1 From an analog signal to bits — Chapter 1

SAMPLING AND RECONSTRUCTION

$f_s > 2W$ (the sampling theorem, W the highest frequency present)

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \operatorname{sinc}\left(\frac{t - nT_s}{T_s}\right)$$

Below the Nyquist rate the replicas of the spectrum overlap and the overlap cannot be undone. $\operatorname{sinc}(x) = \sin(\pi x)/(\pi x)$ throughout this course.

UNIFORM QUANTIZATION

$$\Delta = \frac{2V}{L} = \frac{2V}{2^n}, \quad |e| \leq \frac{\Delta}{2}$$

$$\sigma_q^2 = \frac{\Delta^2}{12}, \quad \text{SQNR}|_{\text{dB}} = 6.02n + 1.76$$

The 1.76 assumes a full-scale sinusoid. The formula is an approximation at small n : at three bits it gives 19.82 dB where the measured value is 19.09 dB, and the gap halves with each extra bit.

RESULT	STATEMENT	CHAPTER
Companding	μ -law and A-law compress before quantizing so that the signal-to-noise ratio is flat across the range	1
PCM rate	$R = nf_s$ bits a second for n bits a sample	1

A.2 Baseband transmission — Chapter 2

THE MATCHED FILTER

$$h(t) = s(T - t), \quad \left(\frac{S}{N} \right)_{\max} = \frac{2E}{N_0}$$

The filter matched to the pulse maximises the signal-to-noise ratio at the sampling instant, and the maximum depends on the energy of the pulse and not on its shape.

BINARY ERROR PROBABILITY AT BASEBAND

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \text{ (antipodal),} \quad P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right) \text{ (on-off, orthogonal)}$$

NYQUIST CRITERION FOR ZERO INTERSYMBOL INTERFERENCE

$$\sum_k P\left(f + \frac{k}{T}\right) = T \quad \text{for all } f$$

$$B = \frac{1 + \alpha}{2T}, \quad 0 \leq \alpha \leq 1$$

The raised cosine meets the criterion for every roll-off α . At $\alpha = 0$ the bandwidth is the Nyquist minimum $1/2T$ and the pulse decays slowly; at $\alpha = 1$ it is twice that and the pulse decays quickly.

A.3 Signal space — Chapter 3

A SIGNAL AS A POINT

$$s_i(t) = \sum_{k=1}^N s_{ik} \psi_k(t), \quad s_{ik} = \int_0^T s_i(t) \psi_k(t) dt$$

$$E_i = \|\mathbf{s}_i\|^2 = \sum_{k=1}^N s_{ik}^2, \quad d_{ij} = \|\mathbf{s}_i - \mathbf{s}_j\|$$

The basis $\{\psi_k\}$ is orthonormal and comes from the Gram–Schmidt procedure applied to the signal set. Energy becomes a squared length and difference becomes a distance, which is the move the rest of the course rests on.

A.4 The optimal receiver — Chapter 4

THE DECISION RULE

$$\hat{s} = \arg \min_i \left\{ \|\mathbf{r} - \mathbf{s}_i\|^2 - N_0 \ln P(\mathbf{s}_i) \right\} \quad (\text{MAP})$$

$$\hat{s} = \arg \min_i \|\mathbf{r} - \mathbf{s}_i\|^2 \quad (\text{ML, equal priors})$$

With equal priors the rule is minimum distance. The equivalent correlation form is to maximise $\mathbf{r} \cdot \mathbf{s}_i - E_i/2$.

ERROR PROBABILITY

$$P(\mathbf{s}_k \rightarrow \mathbf{s}_j) = Q \left(\sqrt{\frac{d_{kj}^2}{2N_0}} \right)$$

$$P_e \leq \frac{1}{M} \sum_k \sum_{j \neq k} Q \left(\sqrt{\frac{d_{kj}^2}{2N_0}} \right), \quad P_e \approx N_{\min} Q \left(\sqrt{\frac{d_{\min}^2}{2N_0}} \right)$$

The first is exact for two points. The second is the union bound, and the third its nearest-neighbour form, where $N_{\min}$ is the *average* number of points at the minimum distance and need not be a whole number. The loose form $P_e \leq (M-1)Q(\cdot)$ needs only M and $d_{\min}$.

A.5 Digital modulation — Chapter 5

SCHEME	$d_{\min}^2$	$N_{\min}$
BPSK	$4E_b$	1
BFSK, BASK	$2E_b$	1
M -PSK	$4E_s \sin^2(\pi/M)$	2
M -PAM	$12E_s/(M^2 - 1)$	$2(M - 1)/M$
M -QAM (square)	$6E_s/(M - 1)$	3 at $M = 16$, 3.5 at $M = 64$
M -FSK	$2E_s$ for every pair	$M - 1$

READING THE TABLE

$$P_e \approx N_{\min} Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right), \quad E_s = (\log_2 M)E_b$$

Convert once between symbol and bit energy and never twice. BFSK and BASK need 3.01 dB more than BPSK for the same error probability; doubling M costs about 6 dB in PAM and about 5.33 dB going from 4-PSK to 8-PSK; and 16-QAM beats 16-PAM by 9.29 dB at the same energy.

A.6 Information theory — Chapter 6

INFORMATION AND ENTROPY

$$I(s_k) = -\log_2 p_k, \quad H(S) = -\sum_{k=1}^K p_k \log_2 p_k$$

$$0 \leq H(S) \leq \log_2 K, \quad H(S^n) = n H(S)$$

The upper bound is reached when all K symbols are equally likely. The extension result holds because the source is memoryless.

SOURCE CODING

$$\bar{L} = \sum_{k=1}^K p_k l_k, \quad \eta = \frac{H(S)}{\bar{L}} \leq 1$$

$$\sum_{k=1}^K 2^{-l_k} \leq 1, \quad H(S) \leq \bar{L} < H(S) + 1$$

The Kraft inequality is necessary but not sufficient for a prefix code. Over the n -th extension the upper bound becomes $H(S) + 1/n$, so $L_n/n \rightarrow H(S)$.

$$\sigma^2 = \sum_{k=1}^K p_k (l_k - \bar{L})^2$$

Huffman is optimal in average length and is not unique. Placing a merged symbol as high as possible on a tie gives the minimum-variance code among the optimal ones.

APPENDIX B

The laboratories

The course has four laboratories, and each one takes a result the chapters derive on paper and asks you to measure it instead. This appendix runs all four. Every curve, cloud and staircase on the pages that follow was computed when this page was drawn — the noise was generated, the filter was applied, the errors were counted — so what you see is the outcome of a run and not a drawing of one.

That matters for one reason above all. A formula tells you what should happen; a measurement tells you what did. Where the two agree you have learnt that the derivation is sound. Where they disagree you have learnt something better: which assumption in the derivation the experiment broke. Each of the four sections below ends on that disagreement, because it is the part of the laboratory worth carrying away.

Your own numbers will not match these. Each laboratory starts from a number your group is given, so the input waveform, the bit pattern and the source statistics all differ from group to group, and the random noise differs from run to run. What must match is the *shape*: the slope of a curve, the size of a gap, the direction a quantity moves when a setting changes. Read the figures for those.

THE FOUR IN ONE LINE EACH

B.1 builds a uniform quantizer and measures the signal-to-quantization-noise ratio it achieves, then encodes the result as a bit stream and draws it as a waveform. **B.2** sends bits through a noisy channel, recovers them with a matched filter, and measures the bit error rate against the formula of Chapter 2. **B.3** does the same for sixteen-point quadrature amplitude modulation and puts the measured symbol error rate beside the union bound of Chapter 4. **B.4** builds a Huffman code for a source and measures how much it compresses.

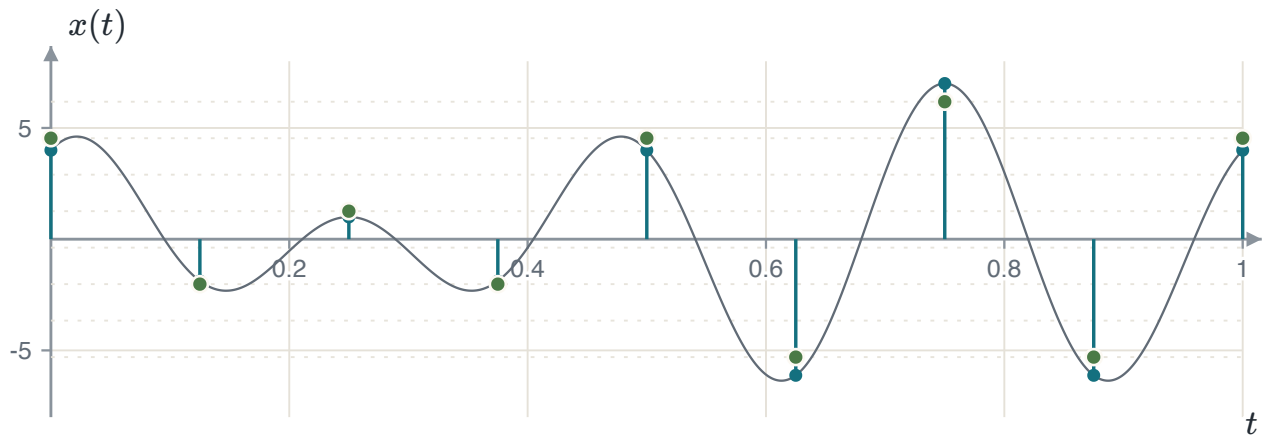
B.1 Quantization, the signal-to-quantization-noise ratio and PCM

The first laboratory walks the whole of Chapter 1 in one sitting. You start with a continuous waveform, sample it at the Nyquist rate, quantize the samples with a uniform quantizer of your own writing, measure the ratio of signal power to quantization-error power, encode each quantized sample as a binary word, and draw the resulting bit stream as a polar NRZ waveform. Then you double the number of levels and do it again.

The waveform is $x(t) = f_1 \sin(2\pi f_1 t) + f_2 \cos(2\pi f_2 t)$, where f_1 and f_2 are two whole numbers your group is given. The case run here has $f_1 = 3$ and $f_2 = 4$, so the highest frequency present is 4 Hz and the Nyquist rate is $f_s = 8$ samples a second. Over the two seconds the laboratory asks for, that is

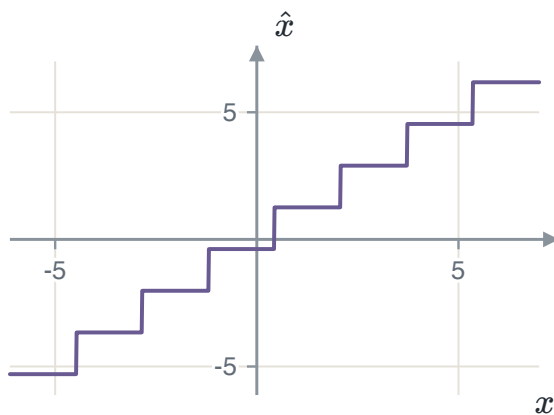
seventeen samples — and seventeen is a smaller number than it looks, which is the point the section ends on.

The quantizer itself is four lines. Find the smallest and largest sample. Divide the distance between them into L equal cells, so the step is $\Delta = (x_{\max} - x_{\min})/L$. Decide which cell each sample falls in. Replace it by the middle of that cell, which is the value that minimises the error inside the cell when nothing is known about where in the cell the sample sits.

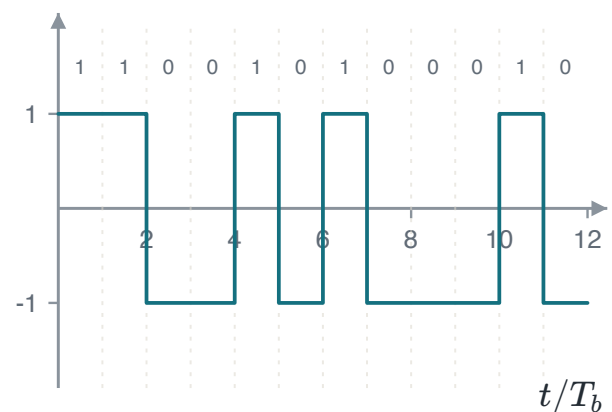


The first second of the run. The pale line is $x(t)$, the stems are the samples x_n taken every $1/8$ s, and the filled circles are the quantized values $\hat{x}_n$ at $L = 8$. The dotted lines are the eight representation levels, spaced $\Delta = 1.640$ V apart. Every circle sits on a dotted line: that is all quantization does.

With $L = 8$ each sample needs $R = \log_2 8 = 3$ bits, and at $f_s = 8$ samples a second the stream leaves the encoder at $R_b = Rf_s = 24$ bit/s. At $L = 16$ it is 32 bit/s. That is the price of the second experiment, and it is worth naming before the reward is measured: a third more bits a second.



The quantizer itself, drawn by feeding it a thousand values across its range. Eight treads, each $\Delta = 1.640$ V wide, each one tread high.

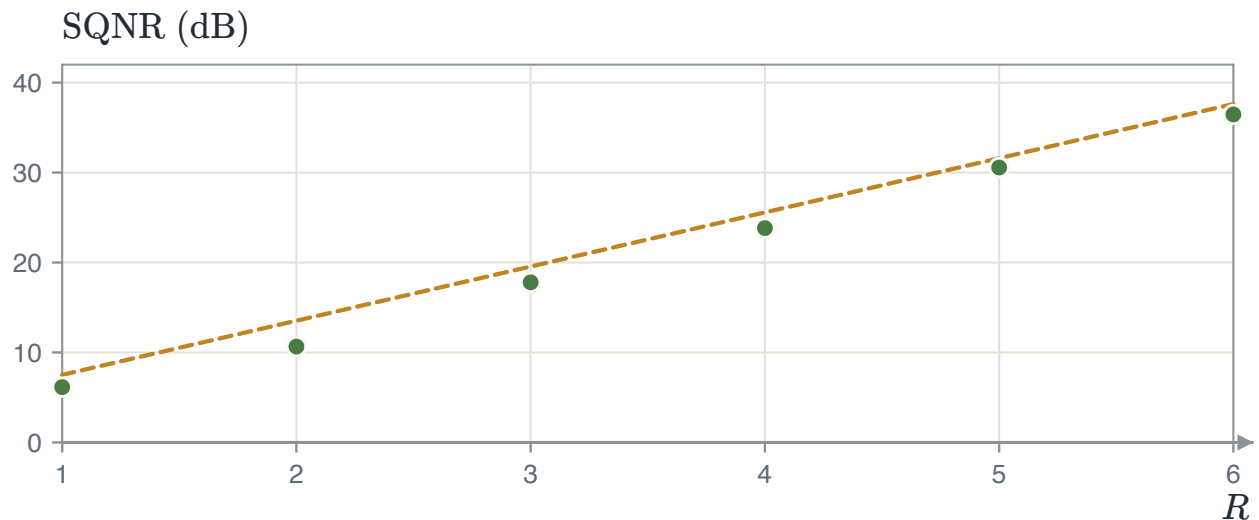


The first four samples encoded and sent. Each sample becomes a three-bit word for its cell, and each bit becomes one bit interval of a polar NRZ waveform: high for a one, low for a zero.

$$\text{SQNR} = \frac{E[X^2]}{E[(X - \hat{X})^2]} \rightarrow \text{SQNR}|_{\text{dB}} = 10 \log_{10} \text{SQNR}$$

Both averages are taken over the samples you actually have. The numerator is the mean square of the samples and the denominator the mean square of the difference between each sample and its quantized value — the same quantity Chapter 1 models as $\Delta^2/12$, but here counted rather than assumed.

On this waveform the measurement gives 17.80 dB at $L = 8$ and 23.83 dB at $L = 16$. The difference is 6.03 dB, which is the six decibels a bit that Section 1.4 predicts, arrived at by counting. That agreement is the main result of the laboratory, and it holds for a reason that has nothing to do with this particular waveform: doubling L halves Δ , halving Δ quarters $\Delta^2/12$, and a factor of four is 6.02 dB.



Measured against modelled, for one to six bits a sample, both computed from the seventeen samples the laboratory takes. The dashed line is the small-step model $10 \log_{10}(P_X/(\Delta^2/12))$ and the circles are the ratio actually measured. The circles rise about six decibels a bit, as they should. But they sit below the line by between 1.0 and 2.9 dB, and the gap neither closes nor settles as R grows. Something is wrong, and the next note says what.

SEVENTEEN SAMPLES IS NOT A POPULATION

The measured points and the modelled line disagree by up to 2.9 dB, and the gap does not shrink as L grows. It is tempting to blame the model, and Section 1.4 gives a real reason it could be blamed — the error is not quite uniform across a cell. But measure the same waveform with two thousand samples instead of seventeen and the gap falls to less than 0.6 dB everywhere. The model was right; the estimate was noisy. Both $E[X^2]$ and $E[(X - \hat{X})^2]$ are averages, and an average over seventeen numbers has a spread of its own. This is the error to name in your report: do not draw a conclusion about a model from a sample too small to test it.

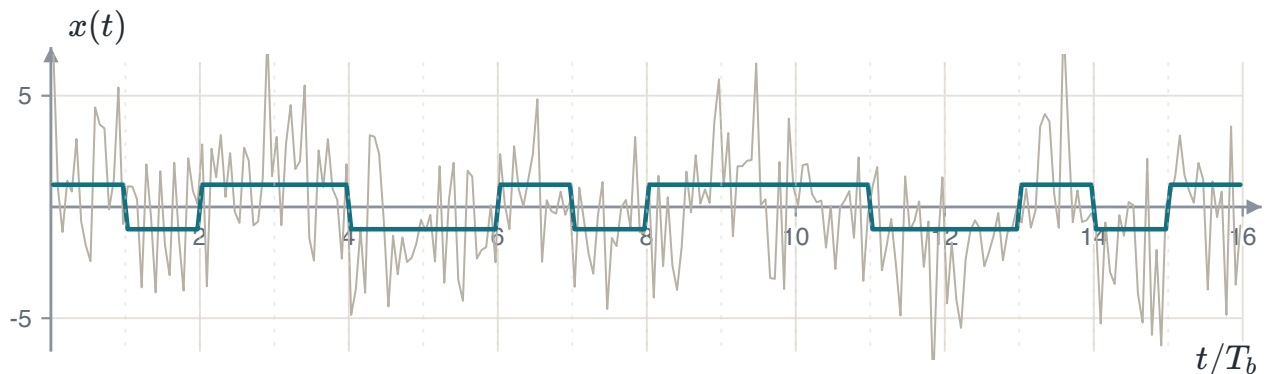
SAMPLING EXACTLY AT THE NYQUIST RATE

The waveform here has a 4 Hz component and the laboratory samples at exactly 8 Hz, which is the boundary the sampling theorem is stated at rather than inside. At exactly twice its frequency, a cosine is sampled at the same two points of every cycle, so that component contributes the same alternating ± 4 to every sample and none of its shape survives. The reconstruction is still correct here because the component is a cosine and lands on its peaks; move the phase and it would not be. Chapter 1 asks for $f_s > 2W$ and means the strict inequality.

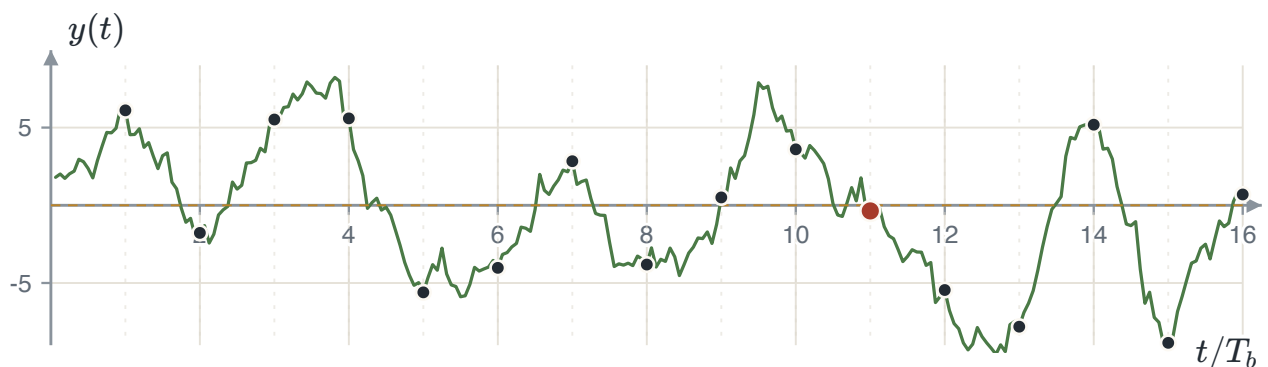
B.2 The matched filter and the bit error rate

The second laboratory builds the receiver of Chapter 2 and measures how often it is wrong. A bit stream becomes a waveform, white Gaussian noise is added to it, a filter matched to the pulse is applied, the filter output is read once a bit, and each reading is compared with a threshold. Then the count of wrong decisions is put beside $Q(\sqrt{2E_b/N_0})$.

The signalling is antipodal. The pulse $\psi(t) = 1/\sqrt{T_b}$ over one bit interval has unit energy, and a bit is sent as $s_1 = A\sqrt{T_b}$ or $s_0 = -A\sqrt{T_b}$ times that pulse — which is a constant $+A$ or $-A$ held for the interval, so the transmitted waveform is the polar NRZ line code of B.1. Its energy a bit is $E_b = A^2T_b$. The noise is added to every sample with variance $N_0/2$, the two-sided density, and the signal-to-noise ratio quoted throughout is $10 \log_{10}(E_b/N_0)$.

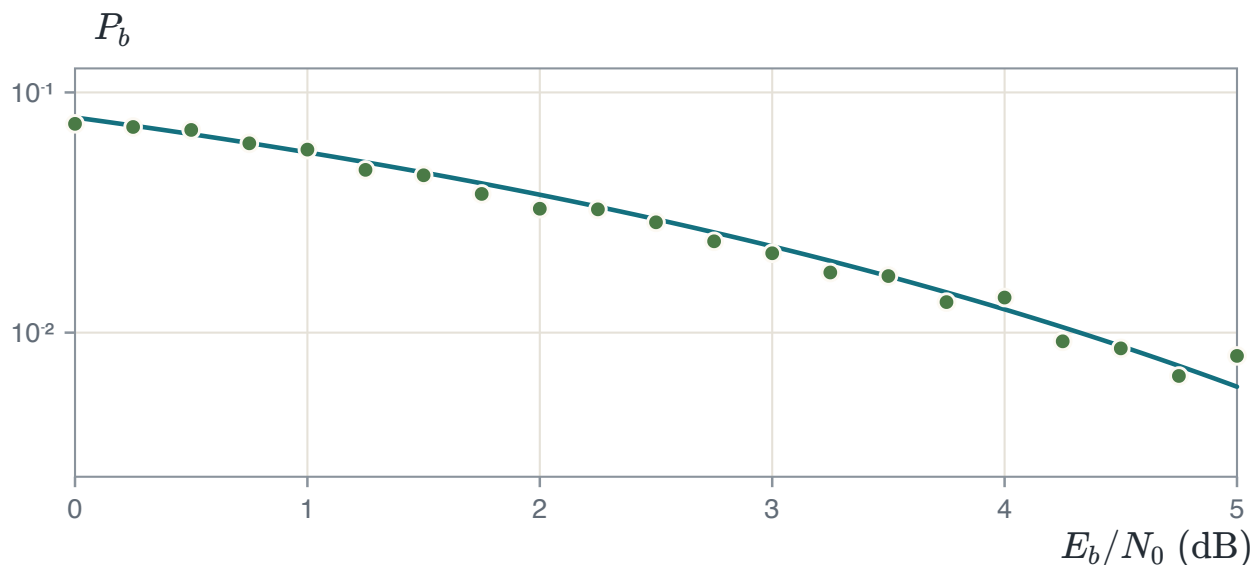


Sixteen bits at $E_b/N_0 = 2$ dB. The clean square wave is the transmitted $s(t)$; the hairline around it is $x(t) = s(t) + w(t)$, which is everything the receiver is given. Nothing about the square wave is visible in the hairline at this noise level, and that is the honest picture — the receiver does not see the signal, it computes with it.



The matched-filter output for the same sixteen bits, read once at the end of every bit interval. The dashed line is the threshold $\lambda = 0$, which is where equal priors put it. Fifteen of the sixteen readings are on the correct side of it. The eleventh is not, and it is drawn in red: at 2 dB the formula expects about one wrong bit in twenty-seven, and this run produced one in sixteen.

One run of sixteen bits settles nothing — one error could as easily have been none or three. The second part of the laboratory therefore drops the picture and repeats the experiment: five thousand bits at each signal-to-noise ratio from 0 to 5 dB in quarter-decibel steps, counting errors and dividing by five thousand.



Measured bit error rate against the formula, five thousand bits a point. The line is $Q(\sqrt{2E_b/N_0})$ and the circles are counted errors. Nowhere on the sweep do the two differ by more than 0.005, and yet the circles clearly wander further from the line towards the right. The next note is about why those two statements are both true.

WHY THE CIRCLES SCATTER, AND WHY THEY SCATTER MORE ON THE RIGHT

Counting k errors in N bits estimates a probability with a spread of about $\sqrt{p(1-p)/N}$. At 0 dB, $p \approx 0.079$, and five thousand bits give a spread of 0.004 — small beside 0.079, so the leftmost circle sits close to the line. At 5 dB, $p \approx 0.006$ and the spread is 0.001, which is a sixth of the value itself: five thousand bits expect about thirty errors there and this run produced forty, so the rightmost circle sits a third above the line. The absolute scatter falls and the relative scatter grows, and on a logarithmic axis it is the relative scatter you see. This is why an error-rate curve needs more bits a point the further right it goes, and why reaching down to 10^{-6} needs millions. Section B.3 does exactly that.

THE THREE-DECIBEL MISTAKE

Two conventions can each cost a factor of two here and neither shows on the page. The first is the noise density: the variance added to each sample is $N_0/2$ because $N_0/2$ is the *two-sided* density, and using N_0 moves every measured point 3 dB to the left. The second is the signalling: $Q(\sqrt{2E_b/N_0})$ is antipodal, where the two waveforms are opposites. If you build on-off signalling by mistake — sending nothing for a zero — the right formula is $Q(\sqrt{E_b/N_0})$ and your points will sit 3 dB to the right of the curve you drew. A measured curve parallel to the theoretical one and 3 dB away from it is almost never noise. It is one of these two.

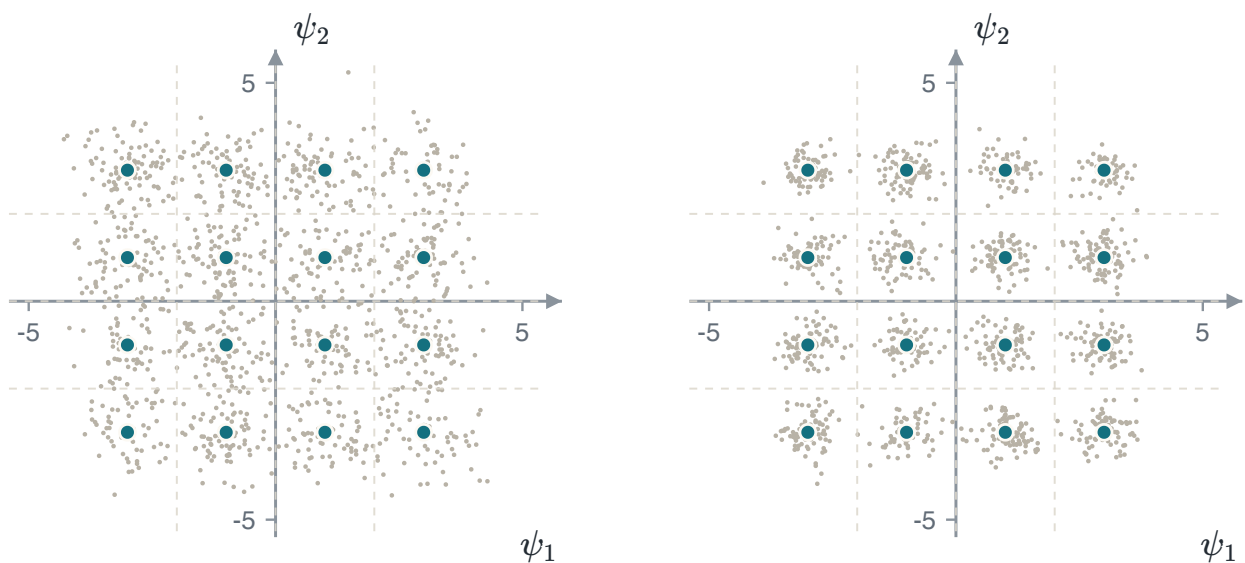
B.3 Sixteen-point QAM and the union bound

The third laboratory moves from one dimension to two. Symbols are drawn from the sixteen-point square constellation of Section 5.4, noise is added independently on each axis, the receiver takes the nearest

point, and the fraction of symbols decided wrongly is put beside the nearest-neighbour approximation of Section 4.4.

Write the sixteen points as $s_I + js_Q$ with each coordinate drawn from $\{-3, -1, 1, 3\}$ in units of half the minimum distance. Then the average symbol energy is $E_s = 10$ in those units, the minimum distance is $d_{\min} = 2$, and since each symbol carries four bits the average energy a bit is $E_b = E_s/4 = 2.5$. Check that against Section 5.4: $d_{\min}^2 = 6E_s/(M-1) = 60/15 = 4$, which is 2^2 . The two routes agree, which is the first thing to verify before generating any noise at all.

The optimal thresholds follow from the same picture. Because the two axes are independent and the levels on each are evenly spaced, the decision boundaries are the three vertical lines and the three horizontal lines midway between neighbouring levels: $\lambda \in \{-2, 0, 2\}$ on each axis. There is nothing to derive beyond the minimum-distance rule of Chapter 4 — in a square grid, nearest point means nearest level on each axis separately.



$E_b/N_0 = 6$ dB. The clouds touch, so a large fraction of the symbols land across a boundary. The measured symbol error rate here is 0.108: about one symbol in nine is decided wrongly.

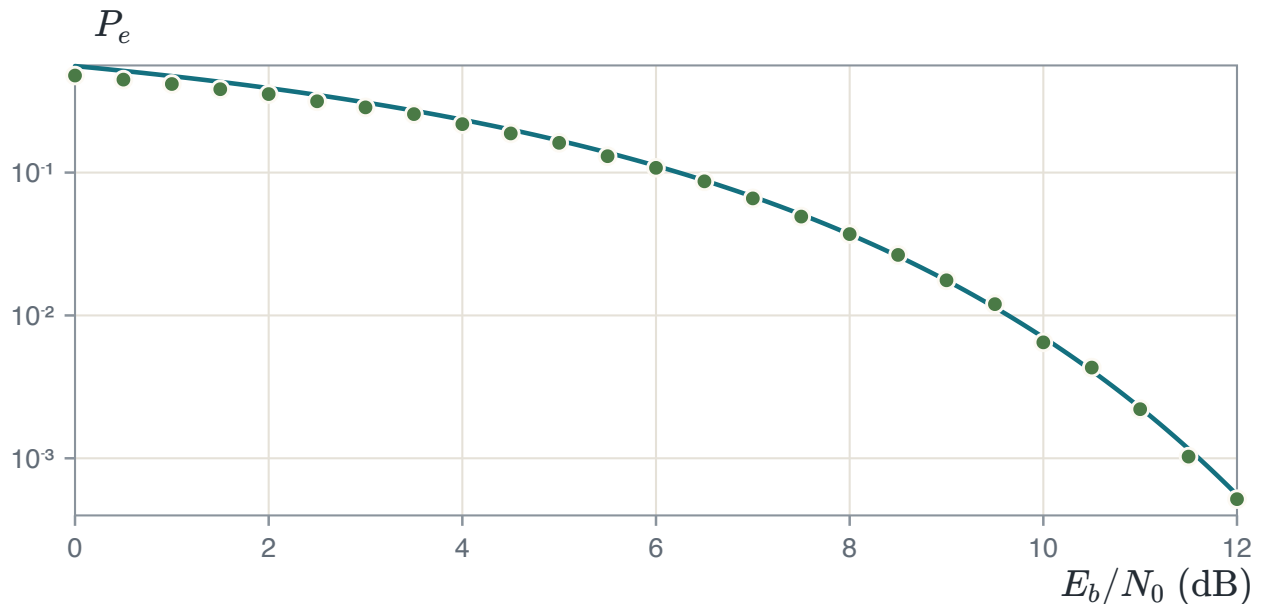
$E_b/N_0 = 11$ dB. Five decibels more energy a bit, and the clouds have pulled apart into sixteen separate blobs. The measured rate falls to 0.0022, about fifty times better.

Twelve hundred received symbols are drawn in each picture so that the clouds can be seen. The rates quoted beneath them do not come from those twelve hundred: a rate of 0.0022 measured on twelve hundred symbols would be three errors, and three is not a measurement. They come from the sweep below, which uses fifty thousand symbols at 6 dB and a quarter of a million at 11 dB. Choosing how many symbols to run is part of the experiment, and Section B.2 said why.

WHAT THE BOUND PREDICTS

$$P_e \approx N_{\min} Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right), \quad N_{\min} = 3, \quad d_{\min}^2 = 4$$

$N_{\min}$ is the average number of nearest neighbours: four corner points have two each, eight edge points three, and four inner points four, and $(4 \cdot 2 + 8 \cdot 3 + 4 \cdot 4)/16 = 3$. It is an average over the constellation and it does not have to be a whole number.



Measured symbol error rate against the nearest-neighbour approximation. The number of symbols is raised as the rate falls — fifty thousand at the left, four hundred thousand at the right — so that every circle rests on a few hundred errors. The circles sit below the line at the left and join it by about 8 dB. That gap is not an error in either the simulation or the formula: it is what the approximation is.

WHY THE MEASUREMENT IS ALLOWED TO BE BETTER THAN THE BOUND

The union bound counts every way a symbol can be mistaken for another and adds the probabilities. Those events overlap — a noise vector large enough to carry a point past two neighbours at once is counted twice — so the sum is larger than the probability of the union, which is why it is a bound and not an equality. At low signal-to-noise ratio the overlaps are common and the bound is loose: here it is 17% above the measurement at 0 dB. At high signal-to-noise ratio a single hop to a nearest neighbour dominates everything else and the overlaps vanish, so the bound tightens onto the truth. From 8 dB on, line and circles agree to within about a tenth either way, and that remaining tenth is the counting spread of a few hundred errors rather than a disagreement — which is why individual circles out there sit above the line as often as below it.

SYMBOL ENERGY AND BIT ENERGY, ONCE

Every quantity in this laboratory can be written per symbol or per bit, and $E_s = 4E_b$ for a sixteen-point constellation. Convert once, at the point where N_0 is computed from the signal-to-noise ratio, and never again. The common failure is to compute N_0 from E_b and then use E_s in the Q function — or the reverse — which moves the whole curve by $10 \log_{10} 4 = 6.02$ dB. A measured curve exactly six decibels from the predicted one, in a laboratory with four bits a symbol, has this and nothing else wrong with it.

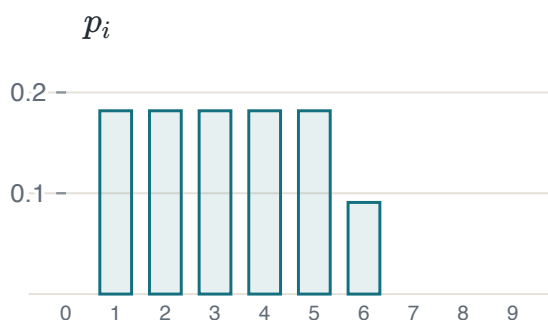
B.4 Huffman coding and how much it compresses

The fourth laboratory leaves the channel behind. A source emits symbols with known probabilities, and the question is how few bits a symbol are needed on average to write down what it emitted, with no loss at all. Chapter 6 gives the floor — the entropy — and gives the algorithm that comes closest to it. Here both are measured.

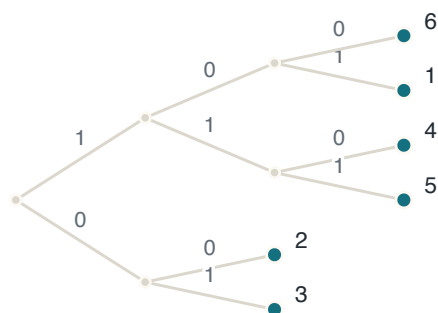
The source alphabet is the ten digits. Your group counts how often each digit occurs in a number it is given and divides by how many digits there are, which gives a probability for each. The case run here uses the eleven digits 1 2 3 4 5 6 5 4 3 2 1: five digits occur twice and one occurs once, so five symbols have probability $2/11$, one has $1/11$, and four have probability zero and take no codeword at all.

DIVIDING BY THE WRONG TOTAL

The probabilities must come from dividing each count by the *number of symbols*. Dividing by anything else — the sum of the digit values, say — gives a set of numbers that does not add up to one, and everything downstream is then quietly wrong: the entropy is not an entropy, the codebook is built for a source that does not exist, and the efficiency can come out above 100%. Print the sum of your probability vector and confirm it is one before you compute anything from it. It is one line, and it catches the only mistake in this laboratory that produces plausible numbers instead of an error message.



The source statistics p_i over the ten-digit alphabet. Four digits do not occur, and a symbol of probability zero is left out of both the entropy and the codebook.



The code the algorithm builds. Reading a path from the root, left is a 0 and right a 1; the digit at each leaf is the symbol that path spells.

The entropy of this source is $H = 2.550$ bits a symbol and the code the algorithm builds has average length $\bar{L} = 2.636$ bits a symbol, so the efficiency is $\eta = H/\bar{L} = 96.7\%$. Two things are worth checking in your report before anything else. The first is that $H \leq \bar{L} < H + 1$, which Section 6.6 guarantees and

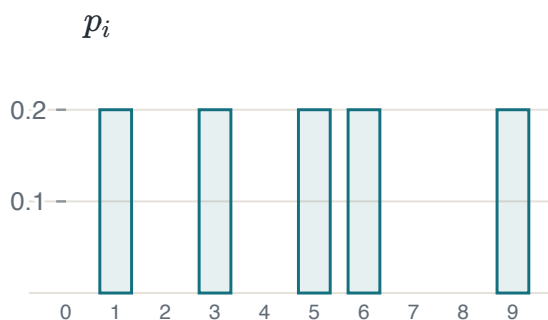
which is here $2.550 \leq 2.636 < 3.550$. The second is that $\bar{L}$ is below the three bits a fixed-length code would need for six symbols, and below the four bits a fixed-length code would need if all ten digits could occur.

A TIE IN THE ALGORITHM IS NOT A MISTAKE

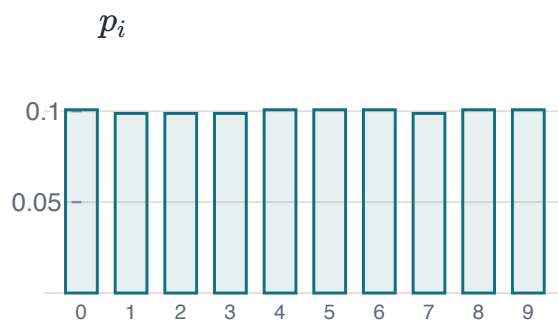
Whenever two nodes have the same probability, the algorithm may merge either, and different choices give different codebooks with the same average length. Your codewords will not match the ones drawn above and need not. What must match is $\bar{L}$, because Huffman is optimal and the optimal average length is unique even when the code is not. If two groups get different average lengths for the same statistics, one of them has an error; if they get different codebooks, neither of them does. Section 6.7 develops the tie-breaking rule that also minimises the variance of the codeword length.

The second part encodes a thousand symbols drawn from the same statistics and compares the length of the result with the length of the same thousand symbols written in fixed-length binary. Ten possible digits need four bits each, so the fixed-length version is 4000 bits; the Huffman version comes to 2621 bits. The compression ratio is $1 - 2621/4000 = 34.5\%$, against the $1 - \bar{L}/4 = 34.1\%$ that the average length alone predicts. The two agree to within half a point, and they are not required to agree exactly: $\bar{L}$ is an average over the distribution, while a thousand draws is a sample of it. If your two numbers differ by much more than a point, the encoder was given the wrong codebook.

The third part asks what the source statistics themselves are worth. Ten counts are drawn from a binomial trial with 50 attempts and success probability ρ , then normalised into a distribution. At $\rho = 0.01$ almost every attempt fails, so most counts come out zero and the surviving symbols are few. At $\rho = 0.99$ almost every attempt succeeds, so all ten counts come out near 50 and the distribution is nearly flat.



$\rho = 0.01$. Five symbols survive and the rest have probability zero. $H = 2.322$ bits, $\bar{L} = 2.400$ bits.



$\rho = 0.99$. All ten symbols occur about equally often. $H = 3.322$ bits, $\bar{L} = 3.395$ bits.

SOURCE	H	$\bar{L}$	η	COMPRESSION AGAINST 4 BITS
The digit source	2.550	2.636	96.7%	34.1%
$\rho = 0.01$	2.322	2.400	96.7%	40.0%
$\rho = 0.99$	3.322	3.395	97.8%	15.1%

WHAT THE THIRD PART IS REALLY ASKING

The efficiency barely moves — 96.7%, 96.7%, 97.8% — while the compression moves from 40% to 15%. Those two numbers measure different things and the laboratory is built to separate them. Efficiency asks how close the code got to the entropy of *its own* source, and Huffman is always close. Compression asks how far below four bits a symbol the source could be pushed at all, and that is set by the entropy, not by the code. At $\rho = 0.99$ the source is nearly uniform over ten symbols, its entropy is 3.322 bits — within a hundredth of $\log_2 10$, the largest it can be — and no lossless code of any kind can do much better than the fixed-length one. There is nothing left to remove.